THE GEOLOGICAL SOCIETY OF AMERICA®

# New Late Cretaceous paleomagnetic results from the Lhasa terrane and their implications for the suturing of India and Eurasia and the closure of the Neo-Tethys Ocean

**Yabo Tong[1,2], Zhenyu Yang[3,†], Junling Pei[1,2], Jianfeng Li[1], Shuchen Jin[3], Lifu Hou[1], Xinxin Sun[3], and Zijian Zhang[3]**
[1]*Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China*
[2]*Key Laboratory of Paleomagnetism and Tectonic Reconstruction, Ministry of Land and Resources, Beijing 100081, China*
[3]*College of Resources, Environment and Tourism, Capital Normal University, Beijing 100048, China*

## ABSTRACT

**The forms of the margins of the Lhasa terrane and the Tethyan Himalaya prior to the collision of India and Eurasia as constrained by paleomagnetism are ambiguous due to the disordered Cretaceous paleomagnetic data from the central Lhasa terrane and the counterclockwise rotation of the Indian plate during the Cretaceous. This ambiguity has induced controversy over the processes of suturing of India and Eurasia and the closure of the Neo-Tethys Ocean. We obtained a set of high-quality Late Cretaceous paleomagnetic data from the central Lhasa terrane, which, integrated with reliable Cretaceous and Paleogene paleomagnetic data sets from the other parts of the Lhasa terrane and Tethyan Himalaya, confirmed that the southern margin of the Lhasa terrane and the northern margin of the Tethyan Himalaya were originally oriented ~317° and ~326°, respectively, prior to the collision of India and Eurasia. The margins of the Lhasa terrane and Tethyan Himalaya were almost consistent with the original straight fold axes of Cretaceous strata in the southern part of the Lhasa terrane, which were oriented 332.5° ± 8.5°, indicating that the subduction of the Neo-Tethys Ocean beneath Eurasia and the movement of the Tethyan Himalaya consistently maintained a stabilized direction of 62.5° ± 8.5° during the Late Cretaceous. The different kinematic characteristics of the Indian plate and Tethyan Himalaya and the overlap of the margins of the Tethyan Himalaya and Lhasa terrane during 59.0–56.0 Ma indicate that the Tethyan Himalaya was already rifted from the Indian plate prior to 62.5–59.2 Ma, and then it quasi-parallelly collided with the Lhasa terrane during 59.0–56.0 Ma, quasi-synchronously closing the Neo-Tethys Ocean.**

## INTRODUCTION

The processes of collision between India and Eurasia remain debated, with interpretations including a single collision between the Indian plate and Eurasia during a wide age range of 65–34 Ma (e.g., Tapponnier et al., 1982; Yin and Harrison, 2000; van Hinsbergen et al., 2011; Chen et al., 2012; Sun et al., 2012; Ma et al., 2014, 2017; Li et al., 2017; Yang et al., 2019); a two-stage continental collision model with a first collision between the Lhasa terrane and the Tethyan Himalaya, and the final collision between the Tethyan Himalaya and India (e.g., van Hinsbergen et al., 2012; Yang et al., 2015a, 2015b; Ma et al., 2016; Yuan et al., 2021); and an arc-continental collision model with a first collision between an island arc and the Lhasa terrane, and the final collision of India with the island arc (Aitchison et al., 2007; Pusok and Stegman, 2020). No matter the model, the processes of the suturing of the Tethyan Himalaya and/or India with Eurasia and the closure of the Neo-Tethys Ocean are still under debate because of the ambiguous nature of the northern margin of the Tethyan Himalaya and the southern margin of the Lhasa terrane during the Late Cretaceous.

Traditional reconstructions for the processes of suturing of the Tethyan Himalaya and/or India with Eurasia and the closure of the Neo-Tethys Ocean have been predominantly based on the assumption that the Lhasa terrane was an E-W extended banded terrane, situated at a latitude in a wide range of ~8.0°N to 25.0°N during the Cretaceous (e.g., Liebke et al., 2010; Chen et al., 2012; Ma et al., 2014; Bian et al., 2017; Cao et al., 2017). In this tectonic setting, the Neo-Tethys Ocean is interpreted to have undergone diachronous closure from west to east (Tapponnier et al., 1981; Besse et al., 1984; Klootwijk et al., 1994; Beck et al., 1995; Rowley, 1996); quasi-synchronous closure along the eastern and western segments of the Yarlung-Zangbo suture zone (Hu et al., 2015, 2016); or diachronous closure from the central Himalaya both eastwards and westwards, respectively (Ding et al., 2017). However, Tong et al. (2019) suggested that the central Lhasa terrane was probably situated at a lower latitude than the other parts of the Lhasa terrane during the Cretaceous, based on the interpretation that the inclinational divergences of the Cretaceous paleomagnetic data sets resulted from as-yet-unidentified Cretaceous syntectonic growth strata in the Lhasa terrane. This would indicate that the Lhasa terrane was not an E-W extended banded terrane during the Cretaceous. Alternatively, Yi et al. (2021) suggested that the lower latitude of the central Lhasa terrane prior to the collision of India and Eurasia probably indicated the southward displacement of the E-W extended banded Lhasa terrane during the Late Cretaceous and Paleocene. This controversy about the paleolatitude and geometry of the Lhasa terrane during the Cretaceous is caused by the disordered paleomagnetic data from the Cretaceous strata in the central Lhasa terrane and the use of unsuitable statistical methods that combined Cretaceous paleomagnetic data sets from different parts of the ~2500-km-long Lhasa terrane together to calculate the average paleolatitude of the whole Lhasa terrane, which resulted in the traditional unsubstantiated assumption of an E-W extended banded Lhasa terrane. Apparently, the average paleolatitude cannot be used to constrain the geometry of the Lhasa terrane in the Cretaceous, because the inhomogeneous crustal deformation of the Lhasa terrane since the collision of India and Eurasia indicates that the original Lhasa terrane would not have been an E-W extended banded terrane (Tong et al.,

†Corresponding author: zhenyu.yang@cnu.edu.cn.

*GSA Bulletin*; November/December 2022; v. 134; no. 11/12; p. 3242–3257; https://doi.org/10.1130/B36310.1; 8 figures; 6 tables.
published online 11 April 2022



For permission to copy, contact editing@geosociety.org
© 2022 Geological Society of America

Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

2019). In addition, the northern margin of the Tethyan Himalaya prior to the collision of India and Eurasia cannot be constrained by the Cretaceous paleomagnetic data sets of the Tethyan Himalaya because of the rapid northward counterclockwise rotational movement of the Tethyan Himalaya during the Cretaceous. Thus, reliable paleomagnetic data sets from the Late Cretaceous strata of the central Lhasa terrane and Paleogene strata of the Tethyan Himalaya are the keys to precisely constraining the processes of suturing of India and Eurasia and closure of the Neo-Tethyan Ocean by comparison of paleolatitudes at different reference points and/or the forms of the margins of the Tethyan Himalaya and Lhasa terrane during the Late Cretaceous.

In order to resolve the foregoing issues, we obtained a set of high-quality Late Cretaceous paleomagnetic data from the central Lhasa terrane, which, integrated with reliable Cretaceous paleomagnetic data sets from other parts of the Lhasa terrane (Fig. 1A) and Paleocene and Eocene paleomagnetic data sets from the Tethyan Himalaya, constrain the forms of the margins of the Lhasa terrane and Tethyan Himalaya prior to the collision of India and Eurasia. We discuss the form of original fold axes of Cretaceous strata in the southern part of the Lhasa terrane, related to the subduction of the Neo-Tethys Ocean beneath Eurasia and the collision of India and Eurasia. The processes of suturing of Indian and Eurasia and closure of Neo-Tethys Ocean are discussed based on the correlation between the margins of the Lhasa terrane and Tethyan Himalaya, and the form of the original fold axes of Cretaceous strata in the southern part of the Lhasa terrane.

## REGIONAL GEOLOGY AND SAMPLING

The Late Cretaceous Shexing volcano-sedimentary sequence of the Lhasa terrane is related to the northward subduction of Neo-Tethys oceanic lithosphere beneath Eurasia (Fig. 1A; Mo et al., 2007; Zhu et al., 2018). In the Maxiang area, in the central Lhasa terrane, the Shexing volcano-sedimentary sequence conformably overlies the Takena Formation, and it is unconformably overlain by the Paleogene Linzizong volcano-sedimentary sequence, which has a basal age of 65.0 Ma (Fig. 1B; Mo et al., 2007). The present work conducted a paleomagnetic study on the upper segment of the Shexing volcano-sedimentary sequence in the Maxiang area. Previous U-Pb zircon dating of andesite lava in the upper segment of the Shexing volcano-sedimentary sequence confirmed the Late Cretaceous age (72.4 ± 1.8 Ma) for the sampled section (Sun et al., 2012).

The sampled strata are mainly composed of basalt, andesite lavas, and interbedded volcanic breccia, red sandstone, siltstone, and mudstone, which experienced tectonic deformation and formed a fold system with WNW-ESE–trending fold axes. Based on the lithologic variation of the volcanic rocks, 41 lava flows were recognized from the sampled section (Fig. 1C). For each lava flow, at least four hand samples were collected to constitute a sampling site. In total, 41 sites (MS1, MS3–MS12, and MS15–MS44) were sampled from basalt and andesite lavas. The bedding attitudes of these sampling sites were precisely measured based on the interbedded volcanic breccias and the red sandstone, siltstone, and mudstone layers. In the laboratory, three to six core samples were drilled from each hand sample. Eight sites (MS13, MS14, MS45–MS50) were obtained from interbedded red sandstone, siltstone, and mudstone, and one site (MS2) was obtained from the matrix of interbedded volcanic breccia. In total, 102 core samples were obtained from these sampling sites using a portable drill and oriented with a magnetic compass. In order to verify that sampled lavas at each site did not affect the magnetic compass readings, the magnetic compass was oriented to 0° at a location ~1 m away from the lavas, and then the magnetic compass was slowly moved toward the lavas without rotating the magnetic compass. The reading of the magnetic compass kept stationary at 0° during this process, indicating that the magnetic compass was not influenced by the lavas at each site. In addition, a sample for U-Pb zircon dating was also collected from andesite lavas on the top of site MS26. The local declination anomaly of the sampled section was corrected using the present geomagnetic field direction calculated using the 2015 International Geomagnetic Reference Field (Thébault, 2015).

## METHODS AND RESULTS

### Rock Magnetic Measurements and Results

In order to characterize the magnetic mineralogy of the sample section, four representative samples (MS13-9 and MS50-8 from red beds and MS23-4 and MS32-3 from andesite lavas) were used for the following sequence of measurements: isothermal remanent magnetization (IRM) component analysis using a cumulative log-Gaussian approach (Kruiver et al., 2001), thermal demagnetization of a three-component IRM (where direct current [DC] fields of 2.5 T, 0.4 T, and 0.12 T were successively applied to the *z*, *x*, and *y* axes of samples, respectively; Lowrie, 1990), and hysteresis loop measurements.

In the hysteresis loop measurements, the applied field was progressively increased to a maximum of 2.0 T, with an increment of 30 mT and an average time of 2 s. The saturation magnetization ($M_s$), saturation remanent magnetization ($M_r$), coercive force ($B_c$), and remanent coercivity ($B_{cr}$) were determined based on the measured hysteresis loops.

IRM acquisition curves of the two samples from interbedded red beds (MS13-9 and MS50-8) were fitted with four IRM components (Fig. 2A). For sample MS13-9, component 1, with $B_{1/2}$ (the applied DC field at which half of the saturation isothermal remanent magnetization [SIRM] is attained) of 63.1 mT, constituted ~13.0% of the SIRM; component 2, with $B_{1/2}$ of 446.7 mT, constituted ~57.0% of the SIRM; component 3, with $B_{1/2}$ of 1584.9 mT, constituted ~28.0% of the SIRM; and component 4, with $B_{1/2}$ of 12.8 mT, constituted only ~2.0% of the SIRM, indicating that it is not significant. For sample MS50-8, component 1, with $B_{1/2}$ of 50.1 mT, constituted ~4.0% of the SIRM; component 2, with $B_{1/2}$ of 478.6 mT, constituted ~74.0% of the SIRM; component 3, with $B_{1/2}$ of 1584.9 mT, constituted ~18.0% of the SIRM; and component 4, with $B_{1/2}$ of 31.6 mT, constituted only ~3.0% of the SIRM, and thus was not significant. During the thermal demagnetization of the three-component IRM of the two red-bed samples, three components decreased slightly within the temperature interval 200–250 °C, indicating that component 3, with the consistent $B_{1/2}$ of 1584.9 mT, is probably goethite (Fig. 2B). The soft component of the two samples decreased to zero within the temperature interval 560–580 °C; in addition, the hard component of sample MS50-8 also showed significant decrease at 580 °C. Thus, component 1 of the two samples, with $B_{1/2}$ of 63.1 mT and 50.1 mT, respectively, is interpreted as magnetite. When the temperature reached 680 °C, the medium and hard components of the two samples consistently decreased to zero, indicating that component 2 of the two samples, with $B_{1/2}$ of 446.7 mT and 478.6 mT, respectively, is coarse-grained detrital hematite. The rectangular hysteresis loops of the two samples from red beds were closed below 1.6 T, but remained unsaturated (Fig. 2C). The coercive force ($B_c$) and remanent coercivity ($B_{cr}$) had ranges of 125–290 mT and 360–430 mT, respectively. The high $B_c$ and $B_{cr}$ values indicate that the dominant magnetic carrier is a high-coercivity component. $B_{cr}/B_c$ had a range of 1.48–2.88, and $M_r/M_s$ had a range of 0.53–0.69 for the two samples, probably indicating the coexistence of hematite and magnetite. Thus, we concluded that hematite is the dominant magnetic carrier for the red beds, and small amounts of magnetite and goethite are also present.

The IRM acquisition curves and thermal demagnetization of a three-component IRM of



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

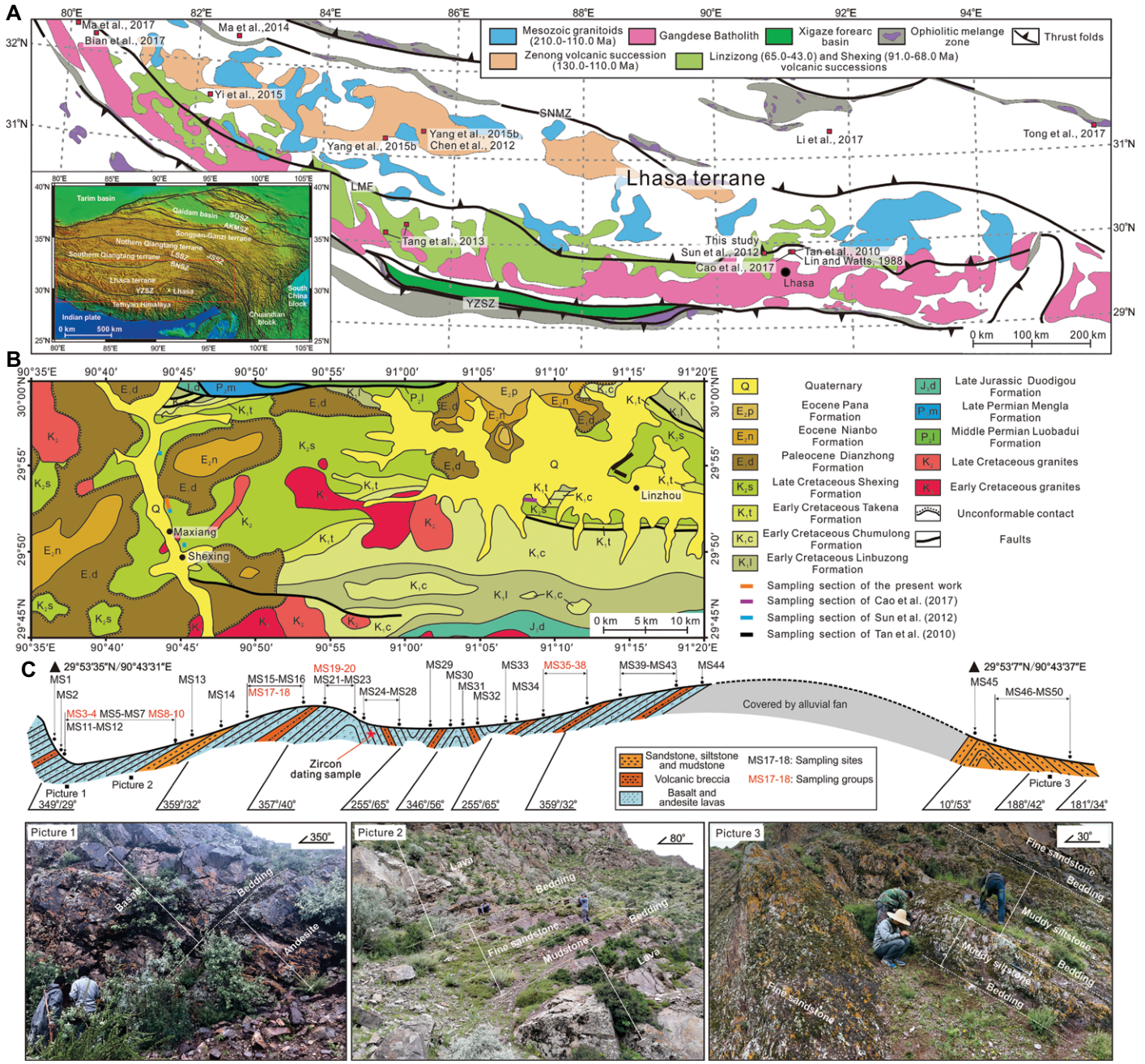



**Figure 1. (A) Schematic tectonic geologic map of the Lhasa terrane and the distributions of paleomagnetic sampling sections in the Cretaceous volcanic rocks and sedimentary strata in the Lhasa terrane. (B) Schematic tectonic geologic map of the Maxiang area and Linzhou Basin, and the distributions of paleomagnetic sampling sections in the Late Cretaceous Shexing Formation in the central Lhasa terrane. (C) Geologic section of the upper segment of the Shexing volcano-sedimentary sequence, showing the locations of sampling sites, lithology, and bedding. SQSZ—South Qilian suture zone; AKMSZ—Ayimaqing-Kunlun-Muztagh suture zone; JSSZ—Jinshajiang suture zone; LSSZ—Longmu Tso–Shuanghu suture zone; BNSZ—Bangong-Nujiang suture zone; YZSZ—Yarlung-Zangbo suture zone; LMF—Luobadui-Milashan fault; SNMZ—Shiquan River–Nam Tso mélange zone.**

the two samples from andesite lavas (MS23-4 and MS32-3) also showed consistent results (Fig. 2). For sample MS23-4, component 1, with $B_{1/2}$ of 446.7 mT, constituted ~39.0% of the SIRM; component 2, with $B_{1/2}$ of 1995.3 mT, constituted ~4% of the SIRM; component 3, with $B_{1/2}$ of 66.1 mT, constituted ~54.0% of the SIRM; and component 4, with $B_{1/2}$ of 14.5 mT, constituted only ~3.0% of the SIRM, and was therefore not significant. For sample MS32-3, component 1, with $B_{1/2}$ of 446.7 mT, constituted ~19.0% of the SIRM; component 2, with $B_{1/2}$ of 1584.9 mT, constituted ~19.0% of the SIRM; component 3, with $B_{1/2}$ of 83.2 mT, constituted ~57.0% of the SIRM; and component 4, with



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

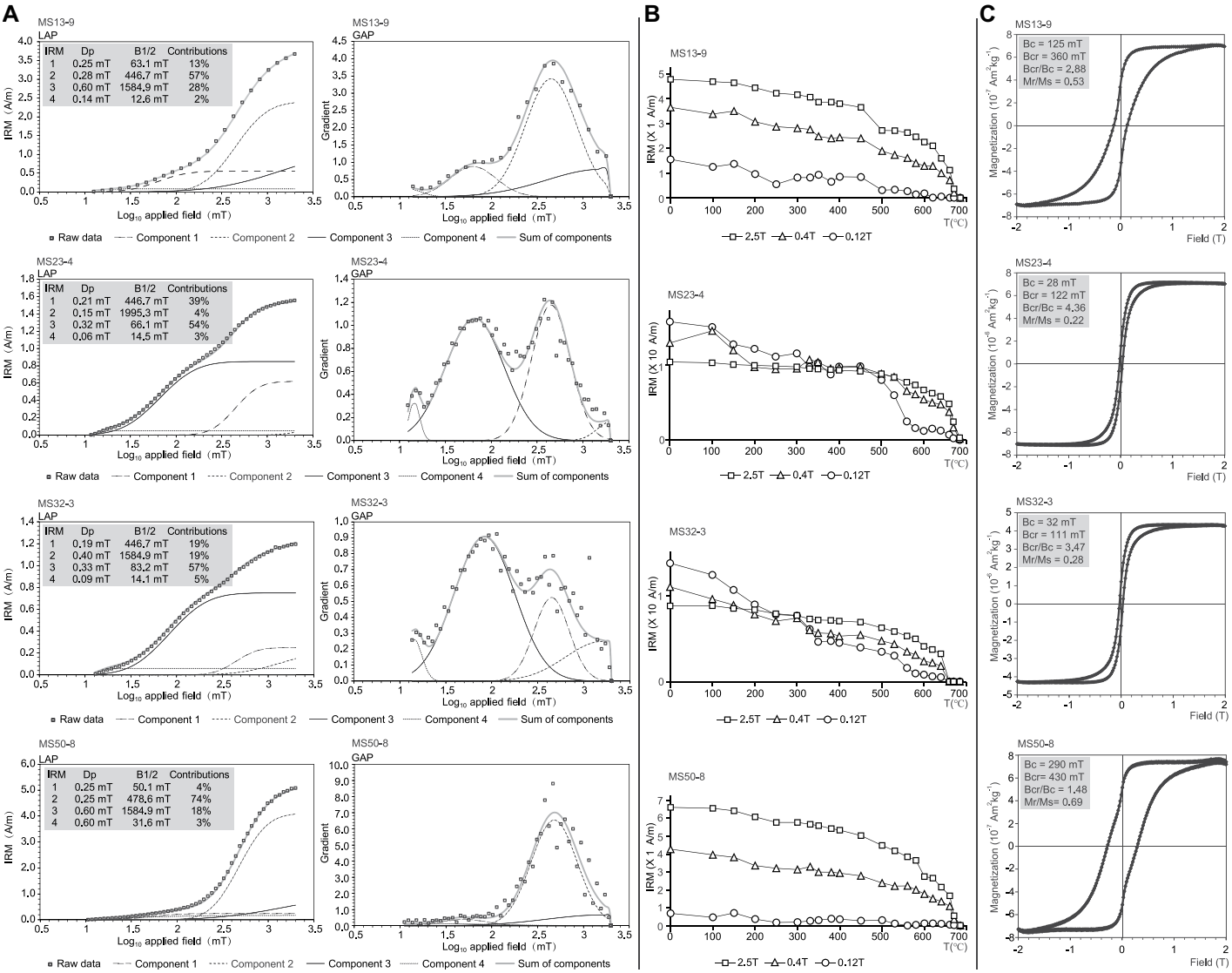



**Figure 2. Rock magnetic results for representative samples. (A) Isothermal remanent magnetization (IRM) component analysis (Kruiver et al., 2001). Components are indicated by lines. $B_{1/2}$—the field at which one half of the saturation isothermal remanent magnetization (SIRM) is reached; Dp—the dispersion parameter, indicating the width of the distribution. (B) Results of thermal demagnetization of a three-component IRM, with direct current (DC) fields of 2.5, 0.4, and 0.12 T applied successively to the *z*, *x*, and *y* axes, respectively (Lowrie, 1990). (C) Hysteresis loops. The applied field was progressively increased to a maximum of 2.0 T, with an increment of 30 mT and an average time of 2s. *Bc*—coercive force; *Bcr*—remanent coercivity; *Ms*—saturation magnetization; *Mr*—saturation remanent magnetization; LAP—The linear acquisition plot; GAP—The gradient of acquisition plot.**

$B_{1/2}$ of 14.1 mT, constituted only ~5.0% of the SIRM, and was therefore not significant. Thermal demagnetization of the three-component IRM revealed slight decreases of the hard and medium components within the temperature interval 200–300 °C, indicating that component 2, with $B_{1/2}$ of 1995.3 mT and 1584.9 mT, respectively, is probably goethite. The soft component of the two samples showed a significant decrease in the temperature interval 540–580 °C; in addition, the hard and medium components also showed a slight decrease within this temperature interval. Thus, component 3, with $B_{1/2}$ of 66.1 mT and 83.2 mT, respectively, is interpreted as magnetite. When the temperature reached 660–680 °C, the three components of the two samples were consistently reduced to zero, indicating that component 1, with a consistent $B_{1/2}$ of 446.7 mT, is hematite. The wasp-waisted hysteresis loops of the two samples were approximately closed at ~600 mT (Fig. 2C). $B_c$ and $B_{cr}$ had ranges of 28–32 mT and 111–122 mT, respectively. The low $B_c$ and $B_{cr}$ values indicate that the dominant magnetic carrier is a low-coercivity component. $B_{cr}/B_c$ had a range of 3.47–4.36, and $M_r/M_s$ had a range of 0.22–0.28 for the two samples, indicating the coexistence of magnetite and hematite/goethite. Thus, we concluded that magnetite is the dominant magnetic carrier for the andesite lavas, and small amounts of hematite and goethite are also present.

## Paleomagnetic Measurements and Results

All core samples were subjected to stepwise thermal demagnetization using an ASC



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

TD-48 thermal demagnetizer. The remanences of samples drilled from sedimentary strata and lava flows were measured using a 2G cryogenic magnetometer and a JR6a spinner magnetometer, respectively, in the Key Laboratory of Paleomagnetism and Tectonic Reconstruction, Ministry of Natural Resources, Beijing, and in the Paleomagnetic Laboratory of Nanjing University. Stepwise thermal demagnetization was performed up to 685 °C for samples from sedimentary strata and andesite lavas, and up to 580 °C for samples from basalt. The initial temperature increment was 50–100 °C within the temperature interval of 20–350 °C, and then the increment was 10–30 °C at the higher temperatures. The thermal demagnetization results for all core samples were plotted on Zijderveld diagrams (Zijderveld, 1967), and the characteristic remanent magnetization (ChRM) was isolated using principal components analysis (Kirschvink, 1980). Fisher statistics (Fisher, 1953) were used to calculate the mean direction for the sampling section.

For the interbedded sedimentary strata, a ChRM could be isolated for sites MS13, MS14, and MS45–MS50 within the temperature interval of 600–685 °C (Fig. 3). A ChRM could be isolated for only four samples from sites MS46–MS49, and therefore they were combined together as an independent site for the application of Fisher statistics (Fisher, 1953). Sites MS13–MS14 have a reversed polarity, and sites MS45, MS46–MS49, and MS50 have a normal polarity (Table 1). The mean paleomagnetic direction of the sedimentary strata is declination ($D_g$) 4.5°, inclination ($I_g$) 23.6°, $k = 3.4$, and $\alpha_{95} = 48.5°$, with $N = 5$ (sampling sites), before tilt correction, and $D_s = 4.6°$, $I_s = 14.6°$, $k = 149.8$, and $\alpha_{95} = 6.3°$ after tilt correction (Fig. 4A). Application of the reversal test (McFadden and Lowes, 1981; McFadden and McElhinny, 1990) produced the following results: $R = 0.31$, $F[4, 2]$ 6.94 at point 5%, critical gamma = 13.7, and critical $R$ at 95% = 1.71, indicating a positive reversal test at the 95% confidence level with classification C.

For the lava flows, the samples from sites MS1, MS2, and MS15 exhibited disordered thermal demagnetization curves, and the ChRM could not be isolated from them. A ChRM could be isolated from the samples from the other sites within the temperature interval of 400–685 °C (Fig. 3). Sites MS3–MS12 and MS26 have a normal polarity, and the other 28 sampling sites have a reversed polarity (Table 1). The ChRM directions of several successive sampling sites were almost identical (e.g., MS3 and MS4, MS8–MS10, MS17 and MS18, MS19 and MS20, and MS35–MS38), indicating that these adjacent lava flows were probably formed in one eruption or over a very short time, and thus the samples from these sites can be considered to have been obtained from the same site. Therefore, they were combined into five sampling groups in order to calculate the group-mean direction. The mean direction of the 26 sampling sites and five groups is $D_g = 9.2°$, $I_g = 41.9°$, $k = 8.7$, and $\alpha_{95} = 9.3°$, with $N = 31$ (sampling sites + groups), before tilt correction, and $D_s = 1.8°$, $I_s = 19.7°$, $k = 21.1$, and $\alpha_{95} = 5.8°$ after tilt correction (Fig. 4B). For the McFadden fold test (McFadden, 1990), $N = 31$ (sampling sites), $\xi_{\text{in situ}} = 13.4 > \xi_c = 6.5 > \xi_{\text{tilt-corrected}} = 1.8$ at the 95% confidence level, and $\xi_{\text{in situ}} = 13.4 > \xi_c = 9.2 > \xi_{\text{tilt-corrected}} = 1.8$ at the 99% confidence level, indicating a positive fold test at the 95% and 99% levels, respectively. The differences between the tilt-corrected mean directions of the lava flows and sedimentary strata were not statistically significant, indicating that the magnetic compass was not influenced by the sampled lavas when measuring the parameters of samples in the field, and the sedimentary strata did not experience significant depositional and compaction-induced inclination shallowing. Thus, the sampled sites and groups from the sedimentary strata and lava flows were combined together for the application of Fisher statistics. The section-mean direction is $D_g = 8.5°$, $I_g = 39.7°$, $k = 7.3$, $\alpha_{95} = 9.5°$, with $N = 36$ (sampling sites), before tilt correction, and $D_s = 2.2°$, $I_s = 19.0°$, $k = 23.8$, and $\alpha_{95} = 5.0°$ after tilt correction (Fig. 4C). The precision parameter ($k$) reached a maximum at 93.5% unfolding during progressive unfolding (Watson and Enkin, 1993), with the direction of $D_{\max} = 2.5°$, $I_{\max} = 20.4°$, $k_{\max} = 24.09$, and $\alpha_{95} = 5.0°$, with $N = 36$, which is consistent with the section-mean direction after tilt correction. In the McFadden fold test (McFadden, 1990), $N = 36$ (sampling sites), $\xi_{\text{in situ}} = 11.6 > \xi_c = 6.9 > \xi_{\text{tilt-corrected}} = 2.8$ at 95% confidence limit, and $\xi_{\text{in situ}} = 11.6 > \xi_c = 9.9 > \xi_{\text{tilt-corrected}} = 2.8$ at 99% confidence level, indicating a positive fold test at the 95% and 99% levels, respectively. Application of the reversal test (McFadden and Lowes, 1981; McFadden and McElhinny, 1990) produced the follow results: $R = 8.1$, $F[20, 48]$ 1.966 at point 5%, and critical gamma = 10.7, indicating a positive reversal test at the 95% confidence level with classification C. Thus, the ChRM obtained from the upper part of the Shexing Formation is of primary geomagnetic origin. Fisher statistics (Fisher, 1953) were conducted on the paleopole of each sampling site and group, giving a Late Cretaceous paleopole of 70.1°N and 265.2°E, with $A_{95} = 4.6°$, for the central Lhasa terrane (Table 1).

### SHRIMP Zircon U-Pb Data

Zircon grains were separated from samples of andesite lavas taken from the top of site MS26 using standard density and magnetic separation methods in the Langfang Laboratory of Geophysical Exploration. Most of the zircon grains were euhedral to subhedral and showed prismatic forms with oscillatory zoning, indicating igneous features (Fig. 5A). Dating of the zircon grains was carried out using the sensitive high-resolution ion microprobe (SHRIMP) II at the Beijing SHRIMP Centre, Chinese Academy of Geological Sciences. Mass resolution during the analytical sessions was ~5000 (1% height). Spot sizes were ~25 μm, and each site was rastered for 150 s before analysis. Data processing was conducted using Squid and Isoplot software (Ludwig, 2003). The SHRIMP U-Pb analyses of zircon grains from the MS26 sampling site are presented in Table 2 and Figure 5. The Th/U values of the zircon grains had a range of 0.34–1.08, and the $^{206}Pb/^{238}U$ ages of the major age group gave a range of ca. 71.54–68.4 Ma (Fig. 5B). Thirty-one analyzed zircon grains defined a concordant age of 70.51 ± 0.31 Ma (mean square of weighted deviates [MSWD] 0.88; Fig. 5C), which represents the age of the andesite from site MS26. Combined with the previous U-Pb zircon date (72.4 ± 1.8 Ma) of andesite lava in the upper segment of the Shexing volcano-sedimentary sequence in the Maxiang area (Sun et al., 2012), we confirmed that the ChRM values obtained from the upper part of the Shexing Formation were acquired at least in the time interval of 74.2–70.20 Ma.

## DISCUSSION

### Southern Margin of the Lhasa Terrane during the Cretaceous

It is generally suggested that the Lhasa terrane was an E-W extended banded terrane, situated at a latitude in a wide range of ~8.0°N to 25.0°N during the Cretaceous (e.g., Liebke et al., 2010; Chen et al., 2012; Ma et al., 2014; Bian et al., 2017; Cao et al., 2017). Alternatively, Yi et al. (2015) suggested that the Lhasa terrane was situated within the paleolatitude range of 14.1°N ± 2.2°N (reference point of 32.3°N, 80.1°E) and ~6°N (reference point of 29.9°N, 91.2°E), and the southern margin of the Lhasa terrane was oriented around 315° in the Late Cretaceous. Tong et al. (2019) suggested a similar reconstruction of the Cretaceous southern margin of the Lhasa terrane, based on the syntectonic sedimentation correction of Cretaceous paleomagnetic data sets of the Lhasa terrane. These results indicated that it is necessary to



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

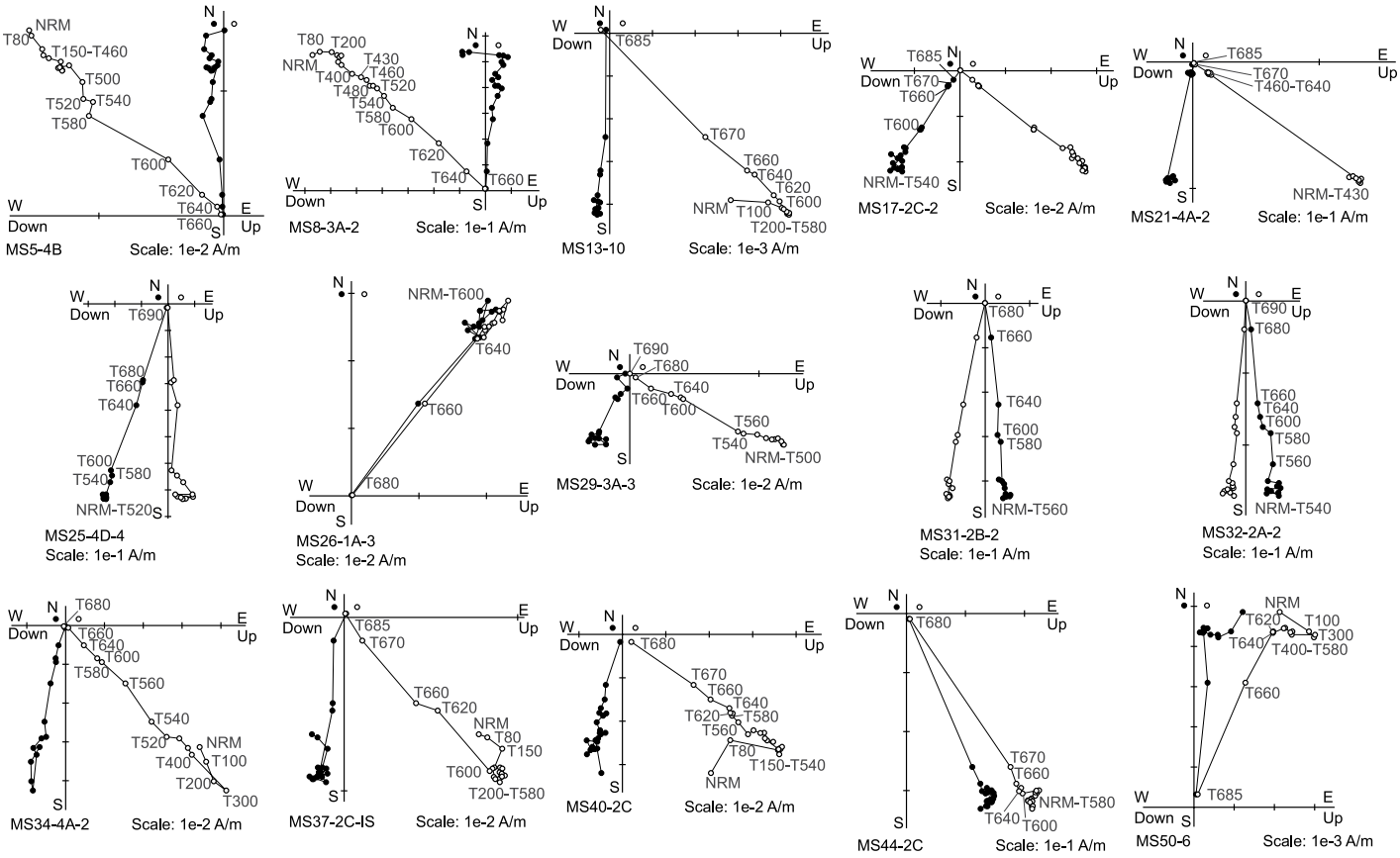



**Figure 3. Orthogonal projections of magnetization vector end points for representative samples. Solid/open symbols are the horizontal/vertical projections in geographic coordinates. NRM—natural remanent magnetization.**

individually constrain the paleolatitude of each part of the Lhasa terrane that currently has an E-W extended banded shape and a total length of ~2500 km. However, the Late Cretaceous paleolatitudes of the central Lhasa terrane (reference point of 29.9°N, 91.2°E) used for reconstruction by Yi et al. (2015) and Tong et al. (2019) were constrained by Paleogene paleomagnetic data sets from the Linzhou Basin under the assumptions that the initial collision of India and Eurasia was later than ca. 50 Ma, and that the inclinational divergences in the Cretaceous paleomagnetic data sets resulted from as-yet-unidentified Cretaceous syntectonic growth strata in the central Lhasa terrane, respectively. Because of the lack of direct constraint on the Late Cretaceous paleolatitude of the central Lhasa terrane by reliable paleomagnetic data sets from volcanic rocks, the previous reconstructions of the southern margin of the Lhasa terrane prior to the collision of India and Eurasia were still defective.

In the present work, high-quality paleomagnetic data obtained from Late Cretaceous volcanic rocks and red beds precisely constrain the central Lhasa terrane to the paleolatitude of 9.8°N ± 2.7°N during 74.2–70.20 Ma. Previous Cretaceous paleomagnetic data sets from the Lhasa terrane that meet the *R* criteria (Meert et al., 2020) are listed in Table 3. We noticed that the paleomagnetic data sets of Tan et al. (2010) from the red beds and lava flows of the Shexing Formation suggested that the Linzhou Basin in the central Lhasa terrane was situated at a paleolatitude of ~24°N in the Late Cretaceous, which is significantly higher than that of the present work and Cao et al. (2017). However, Yi et al. (2015) suggested that the bedding measurements of some sampling sites in the study of Tan et al. (2010) could be wrong, and that local rotational deformation of the sampled strata resulted in significant dispersion of the observed mean declination, which can induce an unsuccessful elongation/inclination (E/I) correction. Thus, the Late Cretaceous paleomagnetic data sets of Tan et al. (2010) were not used for tectonic interpretation in the present work. The syntectonic sedimentation correction was suggested to be necessary to correct the Cretaceous paleomagnetic data sets of the Lhasa terrane (Tong et al., 2019), and this method was then applied in a paleomagnetic study of Late Cretaceous red beds in the western end point of the Lhasa terrane (Bian et al., 2020). However, Zhao et al. (2021) argued that the paleomagnetic data of Bian et al. (2020) displayed a synfolding feature at ~60% unfolding levels, and thus it was probably not suitable for the reconstruction of the Lhasa terrane. The subsequent inspection of Bian and Yang (2021) verified the primary magnetic origin of their paleomagnetic data and confirmed that the Late Cretaceous Jingzhushan Formation red beds could indeed represent syntectonic sedimentation. Considering that syntectonic sedimentation can more reasonably explain the dispersion of inclination in Cretaceous paleomagnetic data sets throughout the Lhasa terrane, the syntectonic sedimentation correction was also used as one of the criteria to filter the Cretaceous paleomagnetic data sets of the Lhasa terrane in the present work. Tang et al. (2013) obtained paleomagnetic data sets from Late Cretaceous lava flows in the western Lhasa terrane. For the lava flows with an age of ca. 99.0 Ma, the $D_s$ of sampling sites in the two limbs of fold are dispersed in a wide range, and thus it can only give a mean $I_s$ of 13.6°. For the lava flows with an age of ca. 93.0 Ma, only one sampling site was in the northeastern limb of fold, and the other sites were



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

TABLE 1. CHARACTERISTIC REMANENT MAGNETIZATION (ChRM) OF THE LATE CRETACEOUS SHEXING VOLCANO-SEDIMENTARY SEQUENC

| Site | Locality (°N, °E) | Bedding strike/dip (°) | *n*/*N* | In situ | | Tilt-corrected | | *k* | $\alpha_{95}$ (°) | VGP | | $A_{95}$ (°) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Dec. (°) | Inc. (°) | Dec. (°) | Inc. (°) | | | Lat. (°N) | Lon. (°E) | |
| MS3* | 29°53′35″, 90°43′31″ | 259/29 | 5/ | 359.4 | 36.6 | 357.4 | 8.0 | 128.8 | 6.8 | – | – | – |
| MS4* | 29°53′35″, 90°43′31″ | 259/29 | 5/ | 0.2 | 39.8 | 357.7 | 11.2 | 468.9 | 3.5 | – | – | – |
| MS3–4 | 29°53′35″, 90°43′31″ | 259/29 | 10/ | 359.8 | 38.2 | 357.6 | 9.6 | 207.0 | 3.4 | 64.9 | 276.3 | 2.4 |
| MS5 | 29°53′35″, 90°43′31″ | 259/29 | 11/ | 7.7 | 45.1 | 2.7 | 17.2 | 66.4 | 5.6 | 68.8 | 263.3 | 4.2 |
| MS6 | 29°53′35″, 90°43′31″ | 259/29 | 9/ | 5.3 | 40.4 | 1.6 | 12.2 | 170.5 | 4.0 | 66.2 | 266.8 | 2.9 |
| MS7 | 29°53′35″, 90°43′31″ | 259/29 | 11/ | 355.9 | 38.9 | 354.5 | 10.0 | 99.0 | 4.6 | 64.6 | 283.6 | 3.4 |
| MS8* | 29°53′35″, 90°43′31″ | 259/29 | 13/ | 1.3 | 41.5 | 358.4 | 13.0 | 171.5 | 3.2 | – | – | – |
| MS9* | 29°53′35″, 90°43′31″ | 259/29 | 13/ | 5.4 | 42.0 | 1.5 | 13.8 | 71.5 | 4.9 | – | – | – |
| MS10* | 29°53′35″, 90°43′31″ | 259/29 | 15/ | 0.1 | 41.7 | 357.5 | 13.1 | 166.9 | 3.0 | – | – | – |
| MS8–10 | 29°53′35″, 90°43′31″ | 259/29 | 41/ | 2.2 | 41.8 | 359.1 | 13.3 | 117.9 | 2.1 | 66.8 | 273.0 | 1.5 |
| MS11 | 29°53′35″, 90°43′31″ | 259/29 | 12/ | 19.2 | 38.5 | 12.8 | 12.5 | 190.6 | 3.2 | 63.5 | 241.1 | 3.0 |
| MS12 | 29°53′35″, 90°43′31″ | 259/29 | 6/ | 354.4 | 41.8 | 353.1 | 12.9 | 147.2 | 5.5 | 65.8 | 287.6 | 4.0 |
| MS13 | 29°53′35″, 90°43′31″ | 269/32 | 10/ | 184.9 | –44.7 | 183.3 | –12.8 | 197.1 | 3.4 | –66.4 | 82.5 | 2.5 |
| MS14 | 29°53′35″, 90°43′31″ | 269/32 | 10/ | 199.5 | –46.5 | 193.5 | –15.9 | 60.3 | 6.3 | –64.8 | 57.8 | 4.6 |
| MS16 | 29°53′32″, 90°43′33″ | 269/32 | 5/ | 198.0 | –47.8 | 192.3 | –17.0 | 23.1 | 16.3 | –65.9 | 59.7 | 12.1 |
| MS17* | 29°53′32″, 90°43′33″ | 269/32 | 19/ | 216.9 | –48.8 | 204.7 | –21.3 | 90.0 | 3.6 | – | – | – |
| MS18* | 29°53′32″, 90°43′33″ | 269/32 | 14/ | 217.0 | –50.5 | 204.1 | –22.9 | 211.9 | 2.7 | – | – | – |
| MS17–18 | 29°53′32″, 90°43′33″ | 269/32 | – | 216.9 | –49.5 | 204.5 | –22.0 | 120.8 | 2.3 | –60.7 | 34.6 | 1.8 |
| MS19* | 29°53′32″, 90°43′33″ | 267/40 | 7/ | 214.0 | –63.2 | 194.7 | –26.9 | 53.2 | 8.4 | – | – | – |
| MS20* | 29°53′32″, 90°43′33″ | 267/40 | 18/ | 208.2 | –64.6 | 191.5 | –27.2 | 76.4 | 4.0 | – | – | – |
| MS19–20 | 29°53′32″, 90°43′33″ | 267/40 | – | 209.9 | –64.2 | 192.4 | –27.1 | 70.2 | 3.5 | –70.7 | 51.7 | 2.8 |
| MS21 | 29°53′28″, 90°43′34″ | 267/40 | 16/ | 196.8 | –62.7 | 186.8 | –23.7 | 84.4 | 4.0 | –71.4 | 69.5 | 3.1 |
| MS22 | 29°53′28″, 90°43′34″ | 267/40 | 14/ | 211.1 | –67.0 | 191.6 | –29.8 | 60.1 | 5.2 | –72.5 | 50.7 | 4.3 |
| MS23 | 29°53′28″, 90°43′34″ | 267/40 | 8/ | 224.9 | –59.4 | 201.9 | –26.1 | 160.1 | 4.4 | –64.1 | 34.5 | 3.5 |
| MS24 | 29°53′28″, 90°43′34″ | 165/65 | 12/ | 194.6 | –14.8 | 163.0 | –32.7 | 73.0 | 5.1 | –70.4 | 146.6 | 4.4 |
| MS25 | 29°53′28″, 90°43′34″ | 165/65 | 12/ | 195.0 | 3.7 | 182.1 | –24.3 | 176.7 | 3.3 | –72.7 | 83.8 | 2.6 |
| MS26 | 29°53′28″, 90°43′34″ | 165/65 | 6/ | 28.2 | –35.2 | 36.9 | 15.3 | 149.2 | 5.5 | 49.0 | 205.7 | 4.0 |
| MS27 | 29°53′28″, 90°43′34″ | 165/65 | 8/ | 202.5 | –23.6 | 156.4 | –42.3 | 45.9 | 8.3 | –68.4 | 171.9 | 8.0 |
| MS28 | 29°53′28″, 90°43′34″ | 165/65 | 6/ | 177.0 | –14.0 | 156.9 | –16.6 | 47.0 | 11.3 | –59.6 | 140.7 | 8.4 |
| MS29 | 29°53′28″, 90°43′34″ | 256/56 | 8/ | 209.0 | –55.1 | 189.1 | –6.4 | 120.2 | 5.1 | –62.0 | 71.1 | 3.6 |
| MS30 | 29°53′28″, 90°43′34″ | 256/56 | 3/ | 204.0 | –62.1 | 183.1 | –10.9 | 51.5 | 17.4 | –65.4 | 83.3 | 12.5 |
| MS31 | 29°53′28″, 90°43′34″ | 165/65 | 15/ | 179.9 | 6.1 | 177.0 | –10.7 | 110.0 | 3.7 | –65.4 | 97.9 | 2.7 |
| MS32 | 29°53′28″, 90°43′34″ | 165/65 | 17/ | 175.1 | –4.1 | 165.5 | –10.9 | 84.5 | 3.9 | –62.0 | 122.8 | 2.8 |
| MS33 | 29°53′28″, 90°43′34″ | 269/32 | 6/ | 188.7 | –36.2 | 186.9 | –4.5 | 463.0 | 3.1 | –61.6 | 76.1 | 2.2 |
| MS34 | 29°53′28″, 90°43′34″ | 269/32 | 10/ | 197.9 | –41.8 | 193.3 | –11.0 | 127.9 | 4.3 | –62.6 | 60.8 | 3.1 |
| MS35* | 29°53′28″, 90°43′34″ | 269/32 | 4/ | 195.1 | –44.8 | 190.7 | –13.7 | 42.6 | 14.2 | – | – | – |
| MS36* | 29°53′28″, 90°43′34″ | 269/32 | 11/ | 200.3 | –43.9 | 194.6 | –13.4 | 98.8 | 4.6 | – | – | – |
| MS37* | 29°53′28″, 90°43′34″ | 269/32 | 10/ | 193.6 | –44.3 | 189.7 | –13.0 | 181.0 | 3.6 | – | – | – |
| MS38* | 29°53′28″, 90°43′34″ | 269/32 | 11/ | 192.9 | –44.5 | 189.1 | –13.2 | 83.3 | 5.0 | – | – | – |
| MS35–38 | 29°53′28″, 90°43′34″ | 269/32 | 36/ | 195.6 | –44.3 | 191.1 | –13.3 | 94.6 | 2.5 | –64.6 | 64.2 | 1.8 |
| MS39 | 29°53′28″, 90°43′34″ | 269/32 | 7/ | 139.0 | –61.7 | 157.6 | –33.7 | 67.0 | 7.4 | –66.6 | 156.5 | 6.3 |
| MS40 | 29°53′28″, 90°43′34″ | 269/32 | 11/ | 193.2 | –48.4 | 188.8 | –17.0 | 89.6 | 4.9 | –67.3 | 67.7 | 3.6 |
| MS41 | 29°53′28″, 90°43′34″ | 269/32 | 9/ | 217.9 | –52.9 | 203.8 | –25.3 | 71.4 | 6.1 | –62.5 | 32.6 | 4.8 |
| MS42 | 29°53′28″, 90°43′34″ | 269/32 | 10/ | 136.6 | –68.6 | 160.2 | –40.3 | 77.3 | 5.5 | –71.0 | 164.1 | 5.1 |
| MS43 | 29°53′28″, 90°43′34″ | 269/32 | 9/ | 138.1 | –60.4 | 156.4 | –32.7 | 304.7 | 3.0 | –65.3 | 156.7 | 2.5 |
| MS44 | 29°53′21″, 90°43′36″ | 269/32 | 8/ | 156.7 | –41.2 | 162.1 | –10.9 | 47.1 | 8.2 | –60.3 | 128.9 | 5.9 |
| MS45 | 29°53′7″, 90°43′37″ | 280/53 | 8/ | 348.0 | 69.4 | 2.0 | 17.6 | 71.6 | 6.6 | 69.0 | 265.2 | 4.9 |
| MS46–49 | 29°53′7″, 90°43′37″ | 98/42 | 4/ | 355.0 | –28.6 | 356.3 | 12.7 | 18.4 | 22.0 | 66.3 | 279.9 | 15.9 |
| MS50 | 29°53′7″, 90°43′37″ | 91/34 | 7/ | 8.2 | –20.3 | 7.9 | 13.5 | 85.5 | 6.6 | 65.8 | 251.3 | 4.8 |
| Site-mean directions of red beds of the Upper Cretaceous Shexing Formation | | | /5 | 4.5 | 23.6 | – | – | 3.4 | 48.5 | – | – | – |
| | | | | – | – | 4.6 | 14.6 | 149.8 | 6.3 | – | – | – |
| Site-mean directions of lava of the Upper Cretaceous Shexing Formation | | | /31 | 9.2 | 41.9 | – | – | 8.7 | 9.3 | – | – | – |
| | | | | – | – | 1.8 | 19.7 | 21.1 | 5.8 | – | – | – |
| Site-mean directions of the Upper Cretaceous Shexing Formation | | | /36 | 8.5 | 39.7 | – | – | 7.3 | 9.5 | 70.1 | 265.2 | 4.6 |
| | | | | – | – | 2.2 | 19.0 | 23.8 | 5.0 | | | |

*Note*: *n*/*N*—number of samples or sampling sites used for Fisher statistic; Dec. and Inc.—Declination and Inclination, respectively; *k*—Fisher precision parameter for samples; $\alpha_{95}$—radius of cone at 95% confidence level of sample-mean direction; VGP—virtual geomagnetic pole; Lat. and Lon.—latitude and longitude of paleopole; $A_{95}$—radius of cone at 95% confidence level about the mean direction. The data with/without the underline means the samples were drilled from sedimentary strata/lavas. (1) McFadden's fold test (McFadden, 1990): $N = 36$, $\xi_{\text{in-situ}} = 11.6 > \xi_c = 6.9 > \xi_{\text{tilt-corrected}} = 2.8$ at 95% confidence limit, and $\xi_{\text{in-situ}} = 11.6 > \xi_c = 9.9 > \xi_{\text{tilt-corrected}} = 2.8$ at 99% confidence limit, indicating a positive fold test at 95% and 99% levels. (2) Reversal test (McFadden and Lowes, 1981; McFadden and McElhinny, 1990): Statistics $R = 8.1$, $F[20,48] = 1.966$, critical gamma = 10.7, positive at 95% confidence limits in classification C.

*Data were not used for Fisher statistic.

all obtained from the southwestern limb of fold. The $I_s$ of this sampling section is 23.0°, which is consistent with that of the two sections of Yang et al. (2015b) ($I_s = 23.4°$ and 25.5°) obtained from the single limb of folds in the adjacent area. Yi et al. (2015) conducted sampling in three sections in the single limbs of Late Cretaceous volcanic rocks in the western end point of the Lhasa terrane. The $I_s$ of the eastern limb of fold



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

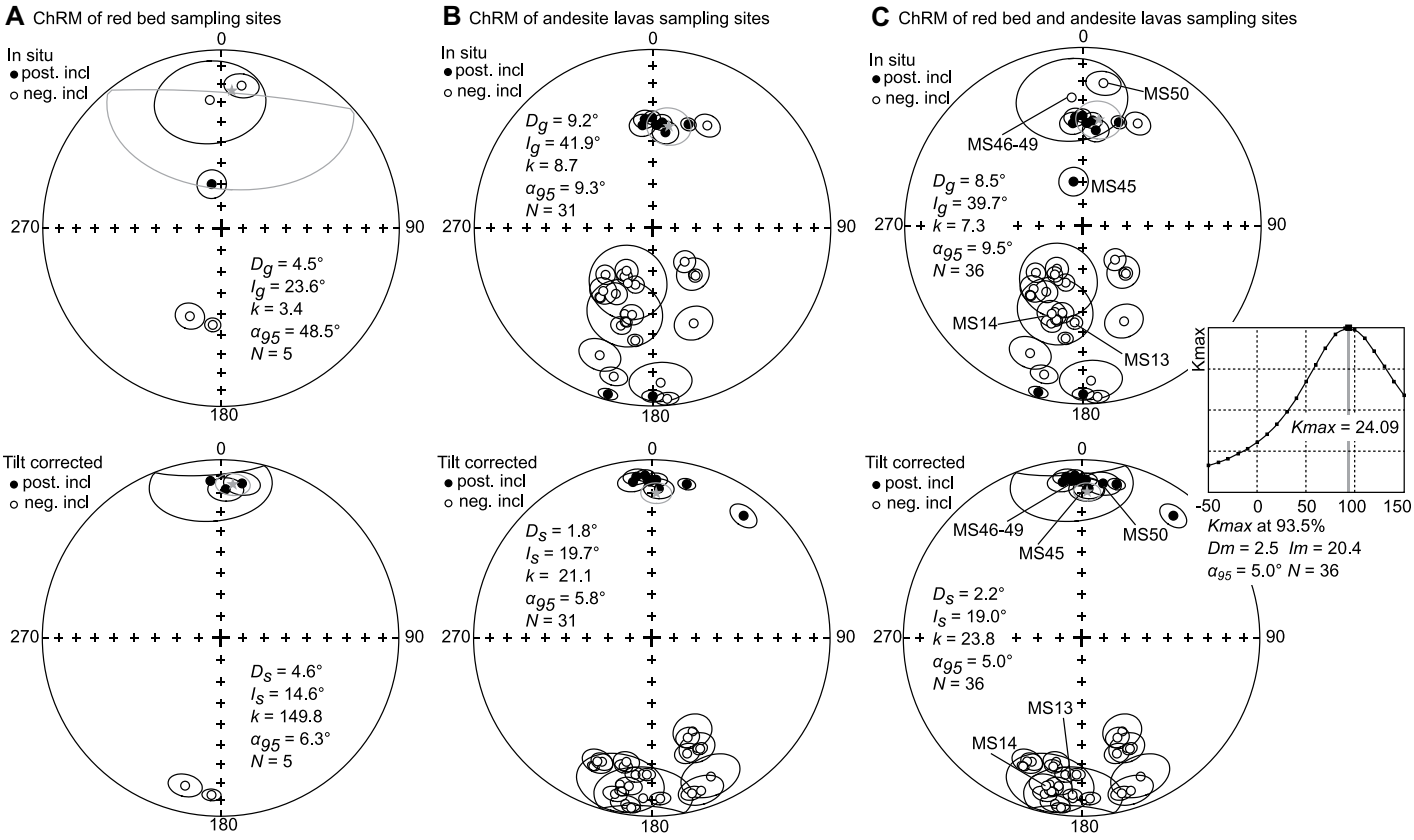



**Figure 4. Equal-area projections of the characteristic remanent magnetization (ChRM). (A) ChRM of sampling sits from red beds. (B) ChRM of sampling sites from lavas. (C) ChRM of sampling sites from red beds and lavas. Stars surrounded by circles denote the 95% confidence limit and the mean direction of the sampling section. *N* is the number of sampling sites used for Fisher statistic.**

is 34.9°, which is significantly higher than that of the sections in the western limb ($I_s = 25.6°$) and the northeastern limb ($I_s = 18.1°$) of folds, but it is consistent with that of the sections of Ma et al. (2018) ($I_s = 34.8°$) and Bian et al. (2017) ($I_s = 36.0°$) in the two limbs of folds in the adjacent area (Table 3). These situations indicate that the original bedding of the sampled volcanic rocks of Yi et al. (2015) in the western limb and the northeastern limb of folds, and the lava flows of Tang et al. (2013) in the southwestern limb of fold were probably not horizontal, because of undulating topography under the lava flows. Thus, the paleomagnetic data sets of Yi et al. (2015) and Tang et al. (2013) were not used for reconstruction.

The filtered Cretaceous paleomagnetic data sets of the Lhasa terrane were obtained from an E-W-extended band with the current latitudinal range of 29.9°N to 31.3°N and longitudinal range of 80.4°E to 95.9°E (Fig. 6A). These paleomagnetic data sets were divided into five groups based on the longitudes of the sampling sections, which respectively corresponded to the five parts of the Lhasa terrane (Table 3; Fig. 6A; Bian et al., 2017, 2020; Cao et al., 2017; Chen et al., 2012; Ma et al., 2014, 2018; Tong et al., 2017; Yang et al., 2015b). Groups 1 (longitude = 95.9°E) and 2 (longitude = 90.7°E–91.1°E) were respectively obtained from the eastern and central Lhasa terrane. Group 3 (longitude = 84.4°E–85.3°E) was obtained from the west-central Lhasa terrane; and groups 4 (longitude = 81.4°E–82.6°E) and 5 (longitude = 80.4°E–80.8°E) were obtained from the western and the western end point of the Lhasa terrane, respectively. The paleomagnetic data for groups 1 and 2 show that the eastern and central Lhasa terrane was situated at a paleolatitude of 12.7°N ± 3.2°N and 9.8°N ± 2.7°N to 10.4°N ± 4.9°N in the Late Cretaceous, respectively. The paleomagnetic data for groups 3, 4, and 5 show that the west-central Lhasa terrane remained at a paleolatitude of 16.2°N ± 2.2°N to 17.1°N ± 2.8°N during 130.0–110.0 Ma and the Late Cretaceous; the western Lhasa terrane remained at a paleolatitude of 19.2°N ± 1.5°N to 19.6°N ± 3.8°N during 132.0–120.0 Ma and the Late Cretaceous; and the western end point of the Lhasa terrane was situated at a paleolatitude of 19.2°N ± 1.4°N to 20.1°N ± 4.8°N during 116.0–72.0 Ma. During the long time span of the Cretaceous, the paleolatitudes of each part of the Lhasa terrane at different times were consistent with each other, indicating that the Lhasa terrane remained stationary (Fig. 6B). Thus, the paleolatitude of each part of the Lhasa terrane was precisely constrained by a small circle fit (Mardia and Gadsden, 1977) to the Cretaceous paleomagnetic data sets from the corresponding part of the Lhasa terrane (Fig. 6C). The results indicate average paleolatitudes of 10.2°N ± 0.3°N for the central Lhasa terrane (reference point of 29.9°N, 91.0°E), 16.6°N ± 0.6°N for the west-central Lhasa terrane (reference point of 31.3°N, 85.0°E), 19.3°N ± 0.4°N for the western Lhasa terrane (reference point of 32.5°N, 82.0°E), and 19.6°N ± 0.9°N for the western end point of the Lhasa terrane (reference point of 32.2°N, 80.6°E; Fig. 6C). The high correlation coefficient ($R = 0.9924$) of the best-fit linear regression for the paleolatitudes indicates that the Lhasa terrane consistently maintained the shape of a triangle with a southward bulge within the paleolatitude range of 19.6°N ± 0.9°N to 10.2°N ± 0.3°N, and the southern margin of



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

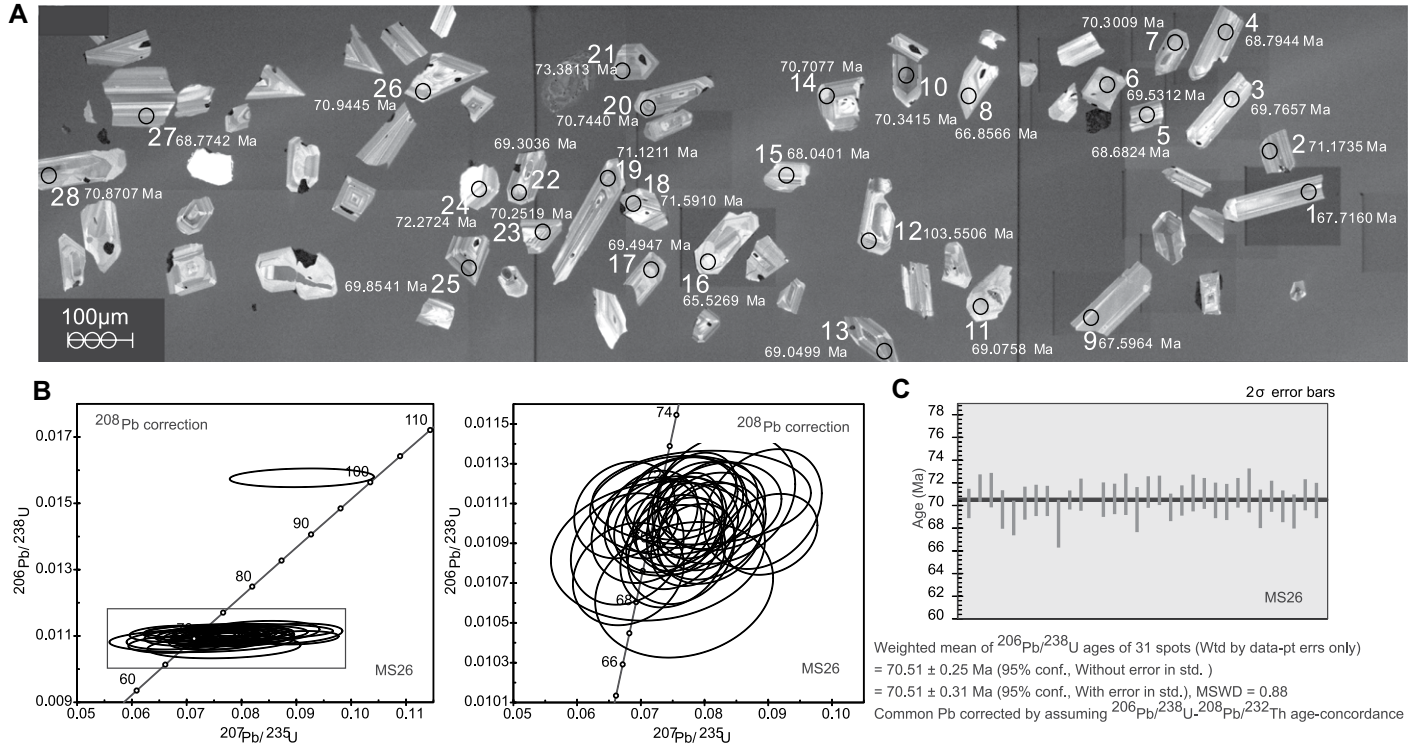



**Figure 5. (A) Cathodoluminescence photographs of zircon grains from andesite at the top of site MS26, and the corresponding $^{206}Pb/^{238}U$ ages of different analytical spots. (B) Concordia plot of the sensitive high-resolution ion microprobe (SHRIMP) U-Pb data for zircons. (C) Weighted average of apparent $^{206}Pb/^{238}U$ ages of 31 spots defining a concordant age of 70.51 ± 0.31 Ma (mean square of weighted deviates [MSWD] = 0.88).**

TABLE 2. SUMMARY OF SENSITIVE HIGH-RESOLUTION ION MICROPROBE (SHRIMP) U-Pb DATA ON ZIRCONS FROM THE TOP OF THE MS26 SAMPLING SITE, SHEXING VOLCANO-SEDIMENTARY SEQUENCE

| Spot | $^{206}pb_c$ (%) | U (ppm) | Th (ppm) | $^{232}Th/^{238}U$ | $^{207}Pb^*/^{206}Pb^*$ | ± (%) | $^{206}Pb^*/^{238}U$ | ± (%) | $^{206}Pb/^{238}U$ age (Ma) | ± | $^{207}Pb/^{206}Pb$ age (Ma) | ± |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MS26-1 | 0.83 | 484 | 383 | 0.82 | 0.0598 | 3.3 | 0.010945 | 0.76 | 70.17 | 0.65 | - | - |
| MS26-2 | 0.49 | 555 | 383 | 0.71 | 0.0512 | 3.1 | 0.011159 | 0.72 | 71.54 | 0.61 | - | - |
| MS26-3 | 2.43 | 370 | 244 | 0.68 | 0.0518 | 8.8 | 0.011130 | 0.87 | 71.35 | 0.76 | 26 | 400 |
| MS26-4 | 0.50 | 293 | 227 | 0.80 | 0.0510 | 4.4 | 0.01086 | 1.0 | 69.65 | 0.85 | - | - |
| MS26-5 | 0.47 | 286 | 136 | 0.49 | 0.0459 | 5.1 | 0.01075 | 0.98 | 68.90 | 0.76 | 69 | 140 |
| MS26-6 | 0.54 | 320 | 130 | 0.42 | 0.0536 | 4.1 | 0.01095 | 0.94 | 70.20 | 0.73 | 158 | 140 |
| MS26-7 | 0.61 | 374 | 165 | 0.46 | 0.0486 | 4.3 | 0.010988 | 0.87 | 70.45 | 0.69 | 92 | 200 |
| MS26-9 | 0.72 | 440 | 237 | 0.56 | 0.0503 | 3.6 | 0.010981 | 0.82 | 70.40 | 0.67 | - | - |
| MS26-8 | 2.03 | 167 | 93 | 0.58 | 0.0522 | 7.3 | 0.01067 | 1.3 | 68.4 | 1.1 | 153 | 590 |
| MS26-10 | 0.33 | 1153 | 686 | 0.62 | 0.0471 | 2.5 | 0.010994 | 0.51 | 70.49 | 0.42 | 58 | 99 |
| MS26-11 | 0.54 | 356 | 144 | 0.42 | 0.0442 | 4.7 | 0.011068 | 0.90 | 70.95 | 0.71 | - | - |
| MS26-12 | 0.74 | 414 | 433 | 1.08 | 0.0418 | 5.9 | 0.01577 | 0.73 | 100.9 | 1.0 | 215 | 150 |
| MS26-13 | 0.48 | 638 | 391 | 0.63 | 0.0507 | 3.0 | 0.01100 | 0.95 | 70.51 | 0.76 | 57 | 120 |
| MS26-14 | 0.57 | 427 | 231 | 0.56 | 0.0501 | 3.9 | 0.011008 | 0.83 | 70.58 | 0.67 | 84 | 140 |
| MS26-15 | 0.35 | 287 | 193 | 0.69 | 0.0584 | 4.1 | 0.01107 | 1.1 | 70.98 | 0.91 | 179 | 140 |
| MS26-16 | 0.85 | 164 | 55 | 0.34 | 0.0485 | 9.3 | 0.01086 | 1.3 | 69.64 | 0.99 | - | - |
| MS26-17 | 1.69 | 395 | 174 | 0.45 | 0.0502 | 4.8 | 0.011105 | 0.84 | 71.19 | 0.69 | 31 | 290 |
| MS26-18 | 1.17 | 500 | 297 | 0.61 | 0.0498 | 4.6 | 0.011126 | 0.76 | 71.33 | 0.65 | 240 | 160 |
| MS26-19 | 0.36 | 496 | 231 | 0.48 | 0.0521 | 3.3 | 0.010892 | 0.78 | 69.83 | 0.60 | 98 | 100 |
| MS26-20 | 0.33 | 408 | 218 | 0.55 | 0.0529 | 3.6 | 0.010987 | 0.83 | 70.44 | 0.66 | 213 | 100 |
| MS26-21 | 1.91 | 347 | 216 | 0.64 | 0.0495 | 7.3 | 0.01109 | 0.92 | 71.10 | 0.81 | 424 | 230 |
| MS26-22 | 1.79 | 412 | 232 | 0.58 | 0.0539 | 4.7 | 0.011085 | 0.82 | 71.07 | 0.68 | 121 | 270 |
| MS26-23 | 2.15 | 321 | 186 | 0.60 | 0.0494 | 5.6 | 0.01099 | 0.93 | 70.45 | 0.78 | - | - |
| MS26-24 | 0.35 | 316 | 168 | 0.55 | 0.0469 | 4.9 | 0.01097 | 0.99 | 70.31 | 0.79 | 70 | 150 |
| MS26-25 | 0.71 | 485 | 245 | 0.52 | 0.0531 | 3.5 | 0.011092 | 0.82 | 71.11 | 0.65 | 46 | 170 |
| MS26-26 | 0.00 | 192 | 94 | 0.51 | 0.0541 | 5.4 | 0.01113 | 1.2 | 71.32 | 0.97 | 451 | 120 |
| MS26-27 | 0.94 | 276 | 158 | 0.59 | 0.0453 | 5.2 | 0.01087 | 1.1 | 69.70 | 0.85 | - | - |
| MS26-28 | 0.78 | 460 | 306 | 0.69 | 0.0472 | 3.9 | 0.011045 | 0.82 | 70.81 | 0.68 | - | - |
| MS26-29 | 0.46 | 365 | 175 | 0.50 | 0.0500 | 4.0 | 0.010906 | 0.89 | 69.93 | 0.70 | 23 | 190 |
| MS26-30 | 0.45 | 292 | 133 | 0.47 | 0.0502 | 4.4 | 0.01084 | 0.97 | 69.47 | 0.75 | 47 | 140 |
| MS26-31 | 0.37 | 395 | 137 | 0.36 | 0.0517 | 3.9 | 0.011066 | 0.89 | 70.94 | 0.68 | - | - |
| MS26-32 | 0.83 | 290 | 119 | 0.43 | 0.0533 | 4.4 | 0.01099 | 1.0 | 70.44 | 0.78 | 54 | 160 |

*Note*: $Pb_c$ and Pb* indicate the common and radiogenic portions, respectively. Common Pb was corrected by assuming $^{206}Pb/^{238}U$-$^{207}Pb/^{235}U$ age concordance.



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

TABLE 3. THE CRETACEOUS PALEOMAGNETIC DATA SETS OF THE LHASA TERRANE THAT MEET THE R CRITERIA (MEERT ET AL., 2020)

| Site coordinates | Time (Ma) | Limbs of fold | *N* | Observed (°) | | | Paleopole (°) | | | Paleolatitude (°N) | IS test | PSV test | Fold and reversal tests | Rock type | Formation | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | $D_s$ | $I_s$ | $\alpha_{95}$ | Plat. | Plon. | $A_{95}$ | | | | | | | |
| Group 1: Paleomagnetic data from the eastern Lhasa terrane | | | | | | | | | | | | | | | | |
| (1) 31.3°N, 95.9°E | K2 | Two limbs | 15 | 0.9 | 24.3 | 5.6 | 71.4 | 273.1 | 5.2 | 12.7 ± 3.2 | A | – | F | Rb | Jingzhushan | Tong et al. (2017) |
| Group 2: Paleomagnetic data from the central Lhasa terrane: average paleolatitude of the two data sets (reference point of 29.9°N, 91.0°E) = 10.2°N ± 0.3°N | | | | | | | | | | | | | | | | |
| (2) 29.9°N, 90.7°E | 74.2–70.2 | Two limbs | 36 | 2.2 | 19.0 | 5.0 | 70.1 | 265.2 | 4.6 | 9.8 ± 2.7 | C | Yes | F, R | V, Rb | Shexing | This study |
| (3) 29.9°N, 91.1°E | 75.0–68.0 | Two limbs | 21 | 0.5 | 20.2 | 6.4 | 70.5 | 269.6 | 4.9 | 10.4 ± 4.9 | C | Yes | F, R | V, Rb | Shexing | Cao et al. (2017) |
| §29.9°N, 91.2°E | 110.0–65.0 | Two limbs | 43 | 350.2 | 42.0 | 2.5 | – | – | – | – | E | Yes | F, R | Rb | Shexing | Tan et al. (2010) |
| Group 3: Paleomagnetic data from the west-central Lhasa terrane: average paleolatitude of the four data sets (reference point of 31.3°N, 85.0°E) = 16.6°N ± 0.6°N | | | | | | | | | | | | | | | | |
| (4)* 31.2°N, 84.4°E | K2 | Southern limb | 11 | 342.8 | 31.6 | 4.5 | 69.0 | 316.3 | 3.7 | 17.1 ± 2.8 | C | – | F | Rb | Jingzhushan | Yang et al. (2015b) |
| (5)* 31.2°N, 84.9°E | K2 | Northern limb | 22 | 302.0 | 30.2 | 3.6 | 35.4 | 353.2 | 3.0 | 16.2 ± 2.2 | C | – | F | Rb | Jingzhushan | Yang et al. (2015b) |
| †31.2°N, 84.9°E | K2 | Northern limb | 22 | 308.3 | 23.4 | 3.6 | – | – | – | – | C | – | F | Rb | Jingzhushan | Yang et al. (2015b) |
| (6)* 31.3°N, 85.3°E | 121.0–117.0 | Northern limb | 12 | 352.7 | 30.5 | 7.7 | 73.7 | 290.8 | 6.4 | 16.4 ± 4.8 | – | Yes | F, R | V | – | Yang et al. (2015b) |
| †31.3°N, 85.3°E | 121.0–117.0 | Northern limb | 12 | 350.3 | 25.5 | 7.7 | – | – | – | – | C | – | F | V | – | Yang et al. (2015b) |
| (7) *31.3°N, 85.1°E | 130.0–110.0 | Southern limb | 17 | 325.2 | 31.0 | 4.3 | 55.2 | 338.5 | 3.5 | 16.7 ± 2.7 | – | Yes | F, R | La, T | Zenong | Chen et al. (2012) |
| †30.2°N, 85.3°E | ca. 93 | Southwestern limb | 6 | 12.2 | 23.0 | 4.0 | – | – | – | – | – | Yes | F | La | – | Tang et al. (2013) |
| †30.9°N, 84.5°E | ca. 99 | Two limbs | 8 | – | 13.6 | 9.4 | – | – | – | – | – | Yes | F | La | – | Tang et al. (2013) |
| Group 4: Paleomagnetic data from the western Lhasa terrane: average paleolatitude of the two data sets (reference point of 32.5°N, 82.0°E) = 19.3°N ± 0.4°N | | | | | | | | | | | | | | | | |
| (8) 32.3°N, 82.6°E | 132.0–120.0 | Two limbs | 51 | 28.2 | 34.5 | 2.3 | 61.4 | 192.9 | 2.1 | 19.2 ± 1.5 | – | Yes | F, R | La | Qushaela | Ma et al. (2014) |
| (9) 32.7°N, 81.4°E | K2 | Two limbs | 54 | 9.5 | 35.5 | 2.6 | 74.4 | 226.0 | 3.8 | 19.6 ± 3.8 | A | Yes | F | Rb | Jingzhushan | Bian et al. (2020) |
| †32.7°N, 81.4°E | K2 | Northern limb | 31 | 13.9 | 19.0 | 3.6 | 51.3 | – | – | – | A | Yes | F | Rb | Jingzhushan | Bian et al. (2020) |
| †32.7°N, 81.4°E | K2 | Southern limb | 23 | 2.7 | 51.8 | 3.1 | 98.5 | – | – | – | A | Yes | F | Rb | Jingzhushan | Bian et al. (2020) |
| †31.6°N, 82.2°E | 79.6 ± 0.7 | Western limb | 15 | 346.6 | 25.6 | 3.5 | 68.4 | – | – | – | – | Yes | F | Ba, Ag | Chalicuo | Yi et al. (2015) |
| Group 5: Paleomagnetic data from the western end point of Lhasa terrane: average paleolatitude of the two data sets (reference point of 32.2°N, 80.6°E) = 19.6°N ± 0.9°N | | | | | | | | | | | | | | | | |
| (10) 32.2°N, 80.8°E | 113.0–72.0 | Two limbs | 38 | 17.3 | 34.8 | 2.1 | 69.7 | 206.6 | 1.9 | 19.2 ± 1.4 | E | – | F, R | Li | Jiega | Ma et al. (2018) |
| (11) 32.2°N, 80.4°E | 116.0–113.0 | Two limbs | 19 | 340.9 | 36.0 | 4.6 | 69.1 | 319.8 | 4.8 | 20.1 ± 4.8 | C | Yes | F, | V, Li | Duoni | Bian et al. (2017) |
| †32.3°N, 80.1°E | 92.4 ± 0.9 | Eastern limb | 5 | 10.0 | 34.9 | 12.3 | 74.2 | 223.8 | 14.0 | – | – | Yes | – | Ba | – | Yi et al. (2015) |
| †32.3°N, 80.1°E | 92.4 ± 0.9 | Northeastern limb | 5 | 30.5 | 18.1 | 7.7 | 53.5 | 202.4 | 6.4 | – | – | Yes | – | Ba | – | Yi et al. (2015) |

*Note*: K2—Upper Cretaceous; *N*—number of sampling sites used for paleomagnetic calculation; $D_s$ and $I_s$—declination and inclination after tilt correction, respectively; Plat. and Plon.—latitude and longitude of paleopole; $\alpha_{95}$ and $A_{95}$—radius of cone at 95% confidence level about the mean direction; IS test—inclination shallowing test; A—anisotropy inclination shallowing test; C—paleomagnetic data of red beds consistent with that of coeval volcanic rocks in the same sampling section; E—elongation/inclination shallowing test; PSV—paleosecular variation; F—positive fold test; R—positive reversal test; Rb—red beds; Ba—basalts; Ag—agglomerates; La—lava flows; Li—limestone; T—tuff; V—volcanic rock.

*Syntectonic sedimentation–corrected paleomagnetic data.

†Syntectonic sedimentation correction was not applied to the data.

§Data were not used for tectonic interpretation.

the Lhasa terrane was oriented around 317° during the Cretaceous (Fig. 6B).

## Subduction Direction of the Neo-Tethys Ocean during the Late Cretaceous

The subduction of the Neo-Tethys oceanic lithosphere beneath Eurasia not only induced the continuous mountain building of an ~2500-km-long, Andean-type Gangdese orogenic belt along the subduction zone, but it also applied continuous stress to southern Eurasia, inducing progressive crustal shortening and deformation of the southeastern part of the Lhasa terrane (Yin and Harrison, 2000; Kapp et al., 2007; Leier et al., 2007). The subduction of the Neo-Tethys Ocean beneath Eurasia along a stabilized direction and the parallel collision of India and Eurasia could have applied a stable stress to the Lhasa terrane, forming a series of straight fold axes in the Cretaceous strata of the Lhasa terrane, which were approximately perpendicular to the subduction direction. Oppositely, changes in the subduction direction of the Neo-Tethys Ocean and diachronous collision of India and Eurasia could have induced a progressive change in the compressional direction, forming a curved tectonic line imparting the fold axes of the Cretaceous strata of the Lhasa terrane (Fig. 7). Thus, the process of suturing of the Neo-Tethys Ocean can be precisely constrained according to the form of the tectonic line represented by the original fold axes of the Cretaceous strata within the Lhasa terrane.

The Cretaceous paleomagnetic data sets from different parts of the Lhasa terrane show significant dispersion of declination, which was related to the different rotational deformation and/or composite folding of sampled strata. In order to reveal the dominant mechanism for the dispersion of declination, the plunge direction and dip of composite fold axes in each section were calculated to correct the Cretaceous paleomagnetic data sets (Table 4). The corrections showed that the observed declinations are indistinguishable from the corrected declinations (Fig. 8A), indicating that the dispersion of declination was not induced by the composite folding. The Cretaceous paleopoles of Europe (Torsvik et al., 2012) were selected as reference poles to constrain the rotational characteristics of sections, and they show that the Lhasa terrane experienced different rotation, with accumulated rotational degrees in a wide range of $-55.2° \pm 6.5°$ to $30.9° \pm 7.3°$ (where the positive and negative values represent clockwise and counterclockwise rotation, respectively). The correlation of Sr–S (where Sr and S are the strike of the reference regional tectonic line and the strike



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

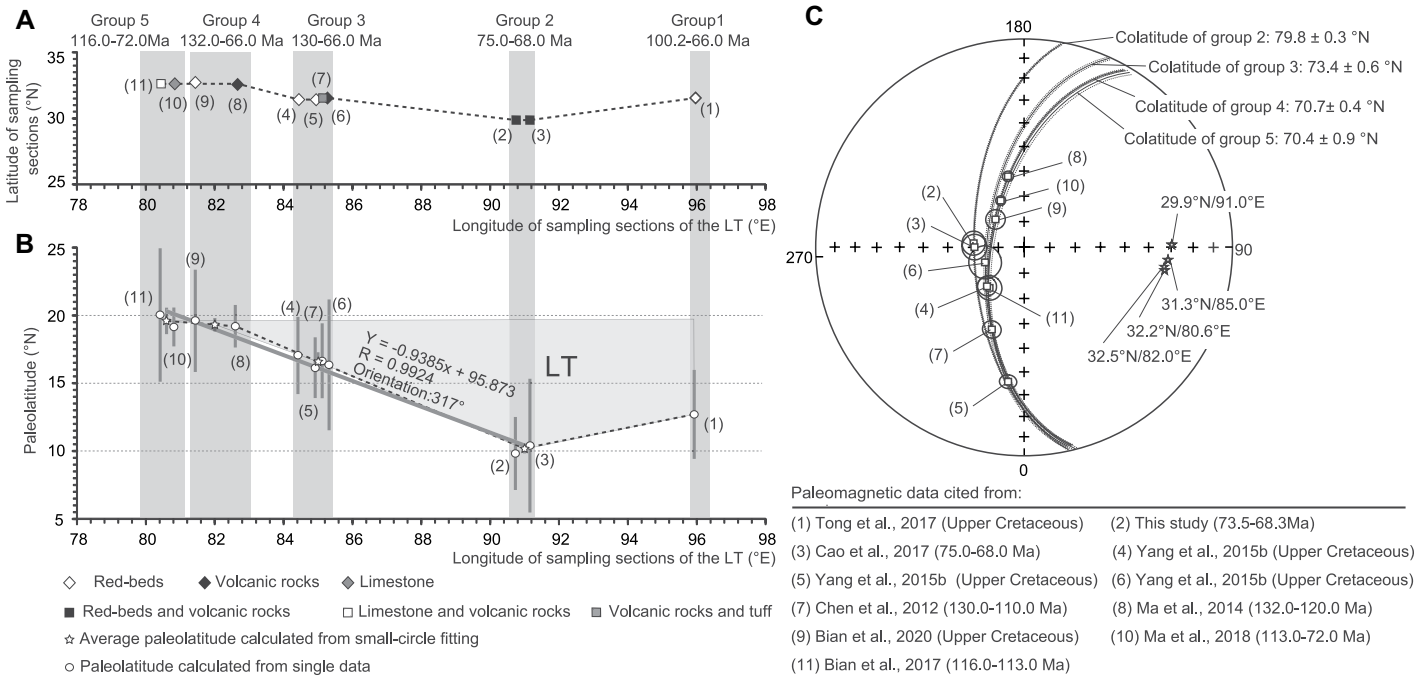



**Figure 6. (A) Longitude of each sampled section of the Cretaceous strata. (B) Observed and average paleolatitude of each sampled section. The average paleolatitudes of groups 2–5 were used to restore the shape of the Lhasa terrane (LT) in the Cretaceous. (C) Small-circle fitting method (Mardia and Gadsden, 1977) used to calculate the average paleolatitudes of groups 2, 3, 4 and 5.**

of the fold axis, respectively) with the accumulated rotational degree (ARD) was used to restore the form of the tectonic line imparting the original fold axes of the Cretaceous strata in the Lhasa terrane. The Sr was assigned the value of 317° based on the southern margin form of the Lhasa terrane. Because the original fold axis can turn to its current orientation by clockwise and counterclockwise rotation, inducing a double solution for Sr–S, the calculation for Sr–S concerns both the rotational characteristics of the sample sections and the principle of selecting the minimum rotational degree (Table 5).

Excluding the data set of Yang et al. (2015b) from the west-central Lhasa terrane, the best-fit linear regression for Sr–S and ARD of the other data sets showed a high correlation coefficient ($R = 0.9234$). When the data sets of Lin and Watts (1988), Li et al. (2017), and the present work from the central Lhasa terrane were also excluded, the $R$ was further increased to 0.9704 (Fig. 8B). The high correlation coefficients indicate that the original fold axes of the Cretaceous strata in the Lhasa terrane composed a straight tectonic line, which was significantly curved by widespread local rotational deformation in the

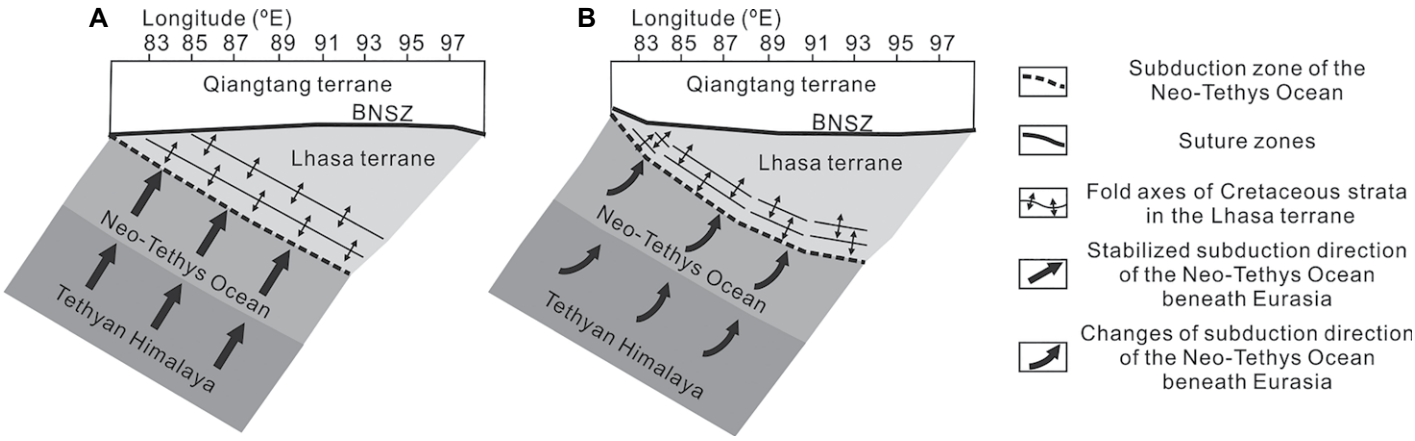



**Figure 7. (A) Subduction of the Neo-Tethys Ocean beneath Eurasia along a stabilized direction could form a series of straight fold axes in the Cretaceous strata that were approximately perpendicular to the subduction direction. (B) Changes of subduction direction of the Neo-Tethys Ocean could result in a curved tectonic line imparted onto the fold axes of Cretaceous strata of the Lhasa terrane. BNSZ—Bangong-Nujiang suture zone.**



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

TABLE 4. TILT CORRECTION OF CRETACEOUS PALEOMAGNETIC DATA OF THE LHASA TERRANE BASED ON THE COMPOSITE FOLD AXES

| Site coordinates | Time (Ma) | *N* | Plunge direction of fold axis (°) | | | Observed (°) | | | Method 1 (°) | | | Method 2 (°) | | | Formation | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | *D* | *I* | $\alpha_{95}$ | $D_s$ | $I_s$ | $\alpha_{95}$ | $D_s$ | $I_s$ | $\alpha_{95}$ | $D_s$ | $I_s$ | $\alpha_{95}$ | | |
| 31.3°N, 95.9°E | K2 | 15 | 84.6 | 0.2 | 1.5 | 0.7 | 24.3 | 6.4 | 0.7 | 24.3 | 6.4 | 0.3 | 24.3 | 7.0 | Jingzhushan | Tong et al. (2017) |
| 29.9°N, 90.7°E | 74.2–70.2 | 36 | 319.2 | 28.6 | 8.5 | 2.2 | 19.0 | 5.0 | 2.4 | 19.0 | 5.0 | 2.0 | 19.1 | 5.1 | Shexing | This work |
| 29.9°N, 91.1°E | 75.0–68.0 | 21 | 270.5 | 25.0 | 5.3 | 0.5 | 20.2 | 6.4 | 0.2 | 20.3 | 6.5 | 358.8 | 20.6 | 7.8 | Shexing | Cao et al. (2017) |
| 29.9°N, 91.2°E | 110.0–65.0 | 43 | 249.9 | 19.2 | 4.0 | 350.2 | 23.5 | 2.5 | 350.3 | 23.5 | 2.6 | 349.8 | 23.6 | 2.7 | Shexing | Tan et al. (2010) |
| 29.9°N, 91.2°E | K2 | 8 | 313.3 | 41.4 | 20.7 | 357.1 | 15.8 | 6.7 | 354.7 | 14.8 | 8.0 | 352.0 | 15.1 | 12.0 | Takena | Lin and Watts (1988) |
| 29.9°N, 90.7°E | 110.0–72.0 | 17 | 75.6 | 6.5 | 11.8 | 344.1 | 34.5 | 5.2 | 344.0 | 34.5 | 5.3 | 345.0 | 33.7 | 5.1 | Shexing | Sun et al. (2012) |
| 31.3°N, 91.9°E | 120.2 ± 0.5 | 19 | 325.1 | 1.2 | 6.2 | 356.4 | 16.4 | 6.3 | 356.3 | 16.4 | 6.3 | 356.1 | 16.2 | 6.7 | Duoni | Li et al. (2017) |
| 31.3°N, 85.1°E | 130.0–110.0 | 17 | 17.5 | 4.6 | 4.7 | 328.3 | 35.9 | 4.4 | 328.3 | 35.9 | 4.4 | 325.2 | 36.9 | 7.3 | Zenong | Chen et al. (2012) |
| 31.3°N, 85.1°E | 121.0–117.0 | 12 | 222.3 | 3.0 | 1.8 | 350.5 | 25.5 | 7.7 | 350.5 | 25.5 | 7.7 | 351.6 | 25.4 | 8.2 | Dianzhong | Yang et al. (2015b) |
| 31.2°N, 84.9°E | K2 | 33 | 280.7 | 3.8 | 7.6 | 316.8 | 30.2 | 5.4 | 316.8 | 30.3 | 5.4 | 316.1 | 30.2 | 5.4 | Jingzhushan | Yang et al. (2015b) |
| 32.3°N, 82.6°E | 132.0–120.0 | 52 | 170.3 | 25.3 | 17.0 | 28.2 | 34.5 | 2.3 | 28.5 | 34.6 | 2.4 | 28.0 | 34.6 | 2.3 | Qushaela | Ma et al. (2014) |
| 32.3°N, 80.1°E | 69.5 ± 2.5 | 17 | 73.8 | 41.1 | 7.3 | 43.3 | 30.3 | 6.9 | 41.2 | 30.1 | 6.2 | 38.0 | 30.9 | 8.6 | Dianzhong | Ma et al. (2017) |
| 32.2°N, 80.4°E | 116.0–113.0 | 13 | 12.0 | 18.2 | 4.0 | 335.6 | 36.2 | 2.8 | 336.3 | 36.3 | 3.3 | 334.9 | 36.5 | 2.8 | Duoni | Bian et al. (2017) |

*Note*: *N*—number of sampling sites used for paleomagnetic calculation; K2—Upper Cretaceous; *D* and *I*—plunging direction and dip of fold axis; $D_s$ and $I_s$—declination and inclination after tilt correction, respectively; Plat. and Plon.—latitude and longitude of paleopole; $\alpha_{95}$—radius of cone at 95% confidence level about the mean direction. Method 1: First rotated the axis of fold back to horizontal and then corrected the data from the residual bedding dip. Method 2: First rotated the bedding strikes in a direction normal to the fold axis direction and then corrected the data from the axis dip.

Cenozoic. The sections of Yang et al. (2015b) in the west-central Lhasa terrane, and those of Lin and Watts (1988), Li et al. (2017), and the present work in the central Lhasa terrane probably experienced insignificant local crustal rotational deformation, and thus their ARDs roughly represent the integral crustal rotation of the west-central and central Lhasa terrane, respectively. These results suggest that the west-central and central Lhasa terrane experienced 55.2° ± 6.5° and 10.6 ± 5.5° to 16.2° ± 6.8° of counterclockwise rotation, respectively, indicating that the current fold axis with a strike of 280.7° in the west-central Lhasa terrane was originally oriented at ~336°, and the fold axes with a strike range of 313.3°–325.1° in the central Lhasa terrane were originally oriented 332.5° ± 8.5°. These results indicate that the Neo-Tethys oceanic lithosphere approximately subducted beneath Eurasia along the stabilized direction of 62.5° ± 8.5° prior to the collision of the Tethyan Himalaya and/or India with Eurasia.

## Process of Suturing of India and Eurasia and Closure of the Neo-Tethys Ocean

The Indian plate experienced conspicuous counterclockwise rotation during northward movement in the Cretaceous, and the northern margin form of the Tethyan Himalaya during the Paleocene was closer to its status at the time of collision with Eurasia. Thus, the paleomagnetic data sets from Paleogene strata of the Tethyan Himalaya are the most qualified materials with which to constrain the northern margin form of the Tethyan Himalaya prior to the initial collision of the Tethyan Himalaya and/or India with Eurasia. Most of the recent Paleocene and Eocene paleomagnetic data sets that meet the *R* criteria (Meert et al., 2020) were obtained from a belt with a current longitude range of 85.3°E to 89.2°E, which corresponds to the west-central and central Lhasa terrane (Table 6).

The paleomagnetic data of Yuan et al. (2021) were obtained from sandstone aged 62.5–59.2 Ma in the Saga area, in the distal part of the Indian plate passive margin, which indicates that the northern margin of the west-central Tethyan Himalaya (reference point of 29.3°N, 85.3°E) was situated at a paleolatitude of 13.7°N ± 2.5°N at that time (Fig. 8C). The other paleomagnetic data sets were obtained from Paleocene and Eocene limestone in the Duina-Gamba area in the southern part of the central Tethyan Himalaya (Yi et al., 2011; Bian et al., 2021), and they show the progressive northward movement, where the southern part of the central Tethyan Himalaya (reference point of 28.3°N, 88.5°E) was situated at a paleolatitude of 6.9°N ± 3.5°N during 62.0–59.0 Ma, and then it moved to 11.2°N ± 2.5°N during 59.0–56.0 Ma, and finally it moved to 15.2°N ± 2.6°N during 51.0–49.5 Ma. The current latitudinal distance between the northern margin of the central Tethyan Himalaya and the Duina-Gamba area is ~120 km. Latitudinal crustal shortening estimation for Indian crustal units in the Himalayan orogeny indicated that ~109–116 km of latitudinal crustal shortening occurred in the Tethyan Himalaya (Parsons et al., 2020). Thus, we estimate that the northern margin of the central Tethyan Himalaya was situated at a paleolatitude of ~8.9°N ± 3.5°N during 62.0–59.0 Ma, and then it moved to ~13.2°N ± 2.5°N during 59.0–56.0 Ma (Fig. 8C). Combined with the paleolatitude of the northern margin of the west-central Tethyan Himalaya (13.7°N ± 2.5°N) during 62.5–59.2 Ma, we suggest that the northern margin of the Tethyan Himalaya in the longitude range of 85.3°E to 89.2°E was originally oriented ~326°, which was not only parallel with the original straight fold axes of Cretaceous strata in the southern Lhasa terrane (332.5° ± 8.5°), but also approximately consistent with the southern margin of the Lhasa terrane (317°). These results indicate that the movement direction of the Tethyan Himalaya during 62.0–59.2 Ma was consistent with the stabilized subduction direction of the Neo-Tethys Ocean (62.5° ± 8.5°) prior to the collision of the Tethyan Himalaya and/or India with Eurasia.

The reconstructions to the Indian plate motions show that it maintained fast northward movement, accompanied by conspicuous counterclockwise rotation since the Early Cretaceous (Besse and Courtillot, 2002; van Hinsbergen et al., 2011; Torsvik et al., 2012). If the Tethyan Himalaya consistently constituted the northern edge of the Indian plate, then the Tethyan Himalaya should also have experienced counterclockwise rotation prior to the collision of India and Eurasia, and thus the movement direction of the Tethyan Himalaya should have undergone significant changes. However, the almost parallel arrangement of the margins of the Lhasa terrane and Tethyan Himalaya and the straight fold axes of the Cretaceous strata in the southern part of the Lhasa terrane indicate that the Tethyan Himalaya maintained northeastward movement along the stabilized direction of 62.5° ± 8.5° during 62.0–59.0 Ma. The different kinematic characteristics of the Tethyan Himalaya and the Indian plate indicate that the Tethyan Himalaya was already no longer a part of the Indian plate at that time. This result is consistent with a multistage collision of India and Eurasia (Yuan et al., 2021; Jadoon et al., 2022), in which the Tethyan Himalaya first rifted from the northern edge of the Indian plate during the Late Cretaceous (probably after ca. 75 Ma), generating the North India Sea (Yuan



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

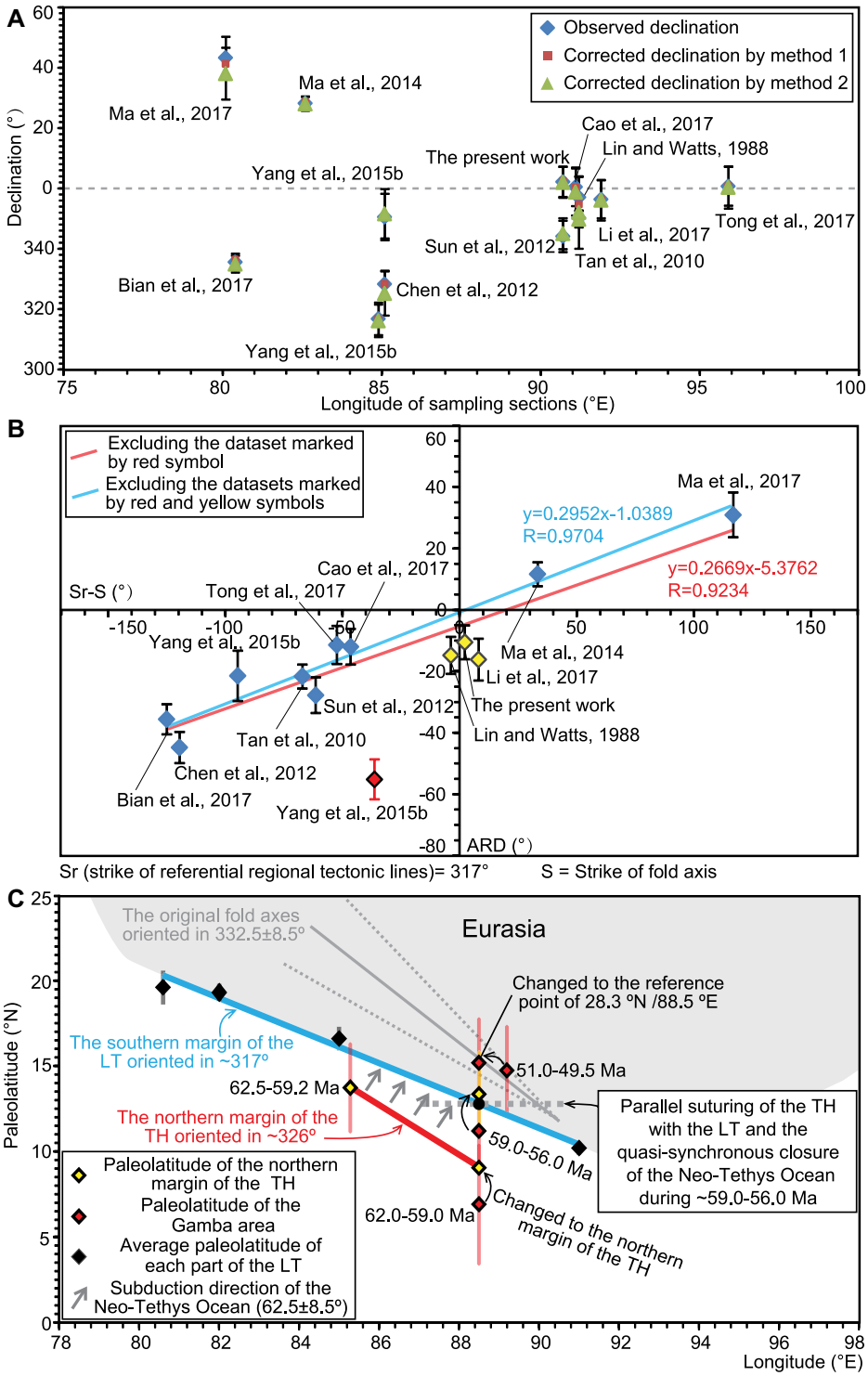



**Figure 8. (A) Observed and tilt-corrected declination by plunge direction and dip of composite fold axis of each section. Method 1: First rotated the fold axis back to horizontal and then corrected the data from the residual bedding dip. Method 2: First rotated the bedding strikes in a direction normal to the fold axis direction and then corrected the data from the axis dip. (B) Best-fit linear regression for Sr (strike of reference regional tectonic lines)–S (strike of fold axis) and the accumulated rotational degree (ARD). (C) Almost-parallel arrangement of the margins of the Lhasa terrane (LT) and Tethyan Himalaya (TH), and the straight fold axes of the Cretaceous strata in the southern part of the Lhasa terrane indicate that the subduction direction of the Neo-Tethys Ocean and the movement of the Tethyan Himalaya were consistently maintained in the stabilized direction of 62.5° ± 8.5° prior to the collision of India and Eurasia. The overlap of the northern margin of the central Tethyan Himalaya with the southern margin of the central Lhasa terrane during 59.0–56.0 Ma indicates that the Tethyan Himalaya quasi-parallelly collided with the Lhasa terrane at that time, which quasi-synchronously closed the Neo-Tethys Ocean.**

et al., 2021). During 59.0–56.0 Ma, the northern margin of the central Tethyan Himalaya began to overlap with the southern margin of the central Lhasa terrane (Fig. 8C), indicating that the Tethyan Himalaya approximately parallelly collided with the Lhasa terrane ca. 59.0–56.0 Ma, which quasi-synchronously closed the Neo-Tethys Ocean along the Yarlung-Zangbo suture zone. This conclusion is supported by the biostratigraphy and detrital zircon geochronology results from Paleogene strata on the northern margin of the Tethyan Himalaya, which show that the Indian plate quasi-synchronously collided with Eurasia along the western and central Tethyan Himalaya (Hu et al., 2015, 2016).

## CONCLUSIONS

The Cretaceous paleomagnetic data sets from the Lhasa terrane and the Paleocene to Eocene paleomagnetic data sets from the Tethyan Himalaya indicate that the southern margin of the Lhasa terrane in the longitude range of 80.6°E to 91.0°E was originally oriented ~317° prior to the collision of India and Eurasia, and the northern margin of the Tethyan Himalaya in the longitude range of 85.3°E to 89.2°E was oriented at ~326° during 62.0–59.2 Ma. The margins of the Lhasa terrane and Tethyan Himalaya were almost parallel with the tectonic line (oriented 332.5° ± 8.5°) recorded by the original fold axes of Cretaceous strata in the southern part of the Lhasa terrane, indicating that the subduction direction of Neo-Tethys oceanic lithosphere beneath Eurasia and the northeastward movement of the Tethyan Himalaya were consistently maintained in a stabilized direction of 62.5° ± 8.5° prior to the collision of the Tethyan Himalaya and Eurasia. The different rotational characteristics of the Indian plate and the Tethyan Himalaya, and the overlap of the northern margin of the central Tethyan Himalaya with the southern margin of the central Lhasa terrane during 59.0–56.0 Ma indicate that the Tethyan Himalaya was already



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

TABLE 5. ROTATIONAL CHARACTERISTICS OF THE LHASA TERRANE

| Site coordinates | Time (Ma) | *N* | Strike of fold axis (°) | Observed (°) | | | Paleopole (°) | | | Rotation (°) | Relative to time (Ma) | Sr-S | IS test | PSV test | Fold test | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | $D_s$ | $I_s$ | $\alpha_{95}$ | Plat. | Plon. | $A_{95}$ | | | | | | | |
| 31.3°N, 95.9°E | K2 | 15 | 84.6 | 0.7 | 24.3 | 6.4 | 71.4 | 273.8 | 5.0 | $-11.4 \pm 6.2$ | 80 | −52.4 | A | – | F | Tong et al. (2017) |
| 29.9°N, 90.7°E | 74.2–70.2 | 36 | 319.2 | 2.2 | 19.0 | 5.0 | 70.1 | 265.2 | 4.6 | $-10.6 \pm 5.5$ | 70 | 2.2 | C | – | F, D | This work |
| 29.9°N, 91.1°E | 75.0–68.0 | 21 | 270.5 | 0.5 | 20.2 | 6.4 | 70.5 | 269.6 | 4.9 | $-12.0 \pm 5.8$ | 70 | −46.5 | C | Yes | F, D | Cao et al. (2017) |
| 29.9°N, 91.2°E | 110.0–65.0 | 43 | 249.9 | 350.2 | 23.5 | 2.5 | 70.2 | 300.5 | 1.9 | $-21.7 \pm 3.9$ | 80 | −67.1 | C | – | F | Tan et al. (2010) |
| 29.9°N, 91.2°E | K2 | 8 | 313.3 | 357.1 | 15.8 | 6.7 | 68.0 | 278.9 | 4.9 | $-14.8 \pm 6.0$ | 80 | −3.7 | C | – | F | Lin and Watts (1988) |
| 29.9°N, 90.7°E | 110.0–72.0 | 17 | 75.6 | 344.1 | 34.5 | 5.2 | 71.9 | 327.2 | 4.5 | $-27.8 \pm 5.8$ | 80 | −61.4 | C | – | F | Sun et al. (2012) |
| 31.3°N, 91.9°E | $120.2 \pm 0.5$ | 19 | 325.1 | 356.4 | 16.4 | 6.3 | 66.9 | 281.2 | 6.1 | $-16.2 \pm 6.8$ | 120 | 8.1 | C | Yes | F, D | Li et al. (2017) |
| 31.3°N, 85.1°E | 130.0–110.0 | 17 | 17.5 | 328.3 | 35.9 | 4.4 | 58.2 | 340.9 | 3.9 | $-44.8 \pm 5.1$ | 120 | −119.5 | – | Yes | F, D | Chen et al. (2012) |
| 31.3°N, 85.1°E | 121.0–117.0 | 12 | 222.3 | 350.5 | 25.5 | 7.7 | 70.1 | 293.2 | 7.4 | $-21.5 \pm 8.2$ | 120 | −94.7 | – | Yes | F, D | Yang et al. (2015b) |
| 31.2°N, 84.9°E | K2 | 33 | 280.7 | 316.8 | 30.2 | 5.4 | 48.0 | 344.3 | 5.3 | $-55.2 \pm 6.5$ | 80 | −36.3 | C | – | F | Yang et al. (2015b) |
| 32.3°N, 82.6°E | 132.0–120.0 | 52 | 170.3 | 28.2 | 34.5 | 2.3 | 61.4 | 192.9 | 2.1 | $11.6 \pm 3.9$ | 130 | 33.3 | – | Yes | F, D | Ma et al. (2014) |
| 32.3°N, 80.1°E | $69.5 \pm 2.5$ | 17 | 73.8 | 43.3 | 30.3 | 6.9 | 47.8 | 181.4 | 6.4 | $30.9 \pm 7.3$ | 70 | 116.8 | C | Yes | F | Ma et al. (2017) |
| 32.2°N, 80.4°E | 116.0–113.0 | 13 | 12.0 | 335.6 | 36.2 | 2.8 | 65.2 | 327.5 | 2.7 | $-35.6 \pm 4.9$ | 100 | −125.0 | C | Yes | F, | Bian et al. (2017) |
| The reference poles of Europe (Torsvik et al., 2012) | | 70 Ma: Plat. 79.2°N, Plon. 175.7°E, $A_{95} = 2.5°$; 80 Ma: Plat. 79.7°N, Plon. 177.9°E, $A_{95} = 2.9°$<br>100 Ma: Plat. 80.8°N, Plon. 152.3°E, $A_{95} = 3.3°$<br>120 Ma: Plat. 79.0°N, Plon. 190.1°E, $A_{95} = 2.6°$; 130 Ma: Plat. 75.0°N, Plon. 183.4°E, $A_{95} = 2.8°$ | | | | | | | | | | | | | | |

*Note*: K2—Upper Cretaceous; *N*—number of sampling sites used for paleomagnetic calculation; $D_s$ and $I_s$—declination and inclination after tilt correction, respectively; Plat. and Plon.—latitude and longitude of paleopole; $\alpha_{95}$ and $A_{95}$—radius of cone at 95% confidence level about the mean direction; Sr—strike of reference regional tectonic lines = 317°; S—strike of fold axis; IS test—inclination shallowing test; C—paleomagnetic data of red beds consistent with that of coeval volcanic rocks in the same sampling section and/or other sampling section; A—anisotropy inclination shallowing test; PSV—paleosecular variation; F—positive fold test; D—dual polarity.

TABLE 6. PRIMARY MAGNETIC COMPONENTS FROM THE LATE CRETACEOUS AND PALEOGENE STRATA OF THE TETHYAN HIMALAYA

| Site coordinates | Time (Ma) | *n*/*N* | Area | Observed (°) | | | | Paleopole (°) | | | Lithology | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | $D_s$ | $I_s$ | $\alpha_{95}$ | Paleolatitude (°N) | Plat. | Plon. | $A_{95}$ | | |
| Jiangzi and Gamba in the southern part of the central Tethyan Himalaya | | | | | | | | | | | | |
| 28.3°N, 88.5°E | 62.0–59.0 | /18 | Gamba | 180.8 | −11.1 | 4.2 | $6.9 \pm 3.5$ | 67.3 | 266.3 | 3.5 | Limestone | Yi et al. (2011) |
| 28.3°N, 88.5°E | 59.0–56.0 | /14 | Gamba | 177.0 | −19.6 | 3.5 | $11.2 \pm 2.5$ | 71.6 | 277.8 | 2.5 | Limestone | Yi et al. (2011) |
| 28.1°N, 89.2°E | 51.0–49.5 | /30 | Duina-Gamba | 332.5 | 27.7 | 2.9 | $14.7 \pm 2.6$ | 61.2 | 337.3 | 2.6 | Limestone | Bian et al. (2021) |
| Paleolatitude of the section of Bian et al. (2021) changed to the reference point of 28.3°N, 88.5°E = 15.2°N ± 2.6°N | | | | | | | | | | | | |
| Saga on the northern margin of the west-central Tethyan Himalaya | | | | | | | | | | | | |
| 29.3°N, 85.3°E | 62.5–59.2 | 86/ | Mubala-Sangdanlin | /176.3 | /–26.0 | 2.6 | $13.7 \pm 2.5$ | 74.0 | 278.5 | 2.5 | Sandstone interbedded with shales and tuffs | Yuan et al. (2021) |

*Note*: *n*/*N*—number samples/sampling sites used for paleomagnetic calculation; $D_s$ and $I_s$—declination and inclination after tilt correction, respectively; Plat. and Plon.—latitude and longitude of paleopole; $\alpha_{95}$ and $A_{95}$—radius of cone at 95% confidence level about the mean direction.

rifted from the northern edge of the Indian plate prior to 62.5–59.2 Ma, and then it quasi-parallelly collided with the Lhasa terrane during 59.0–56.0 Ma. These results support the models of two-stage collision of India with Eurasia and the quasi-synchronous closure of the Neo-Tethys Ocean along the Yarlung-Zangbo suture zone.


## ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (awards 91855216 and 41872221) and the Second Tibetan Plateau Scientific Expedition and Research Program (award 2019QZKK0901). The original paleomagnetic data of the present study are archived online and can be accessed from the public repository at https://doi.org/10.6084/m9.figshare.14882514.


## REFERENCES CITED


Aitchison, J.C., Ali, J.R., and Davis, A.M., 2007, When and where did India and Asia collide?: Journal of Geophysical Research–Solid Earth, v. 112, no. B5, B05423, https://doi.org/10.1029/2006JB004706.

Beck, R.A., Burbank, D.W., Sercombe, W.J., Riley, G.W., Barndt, J.K., Berry, J.R., Afzal, J., Khan, A.M., Jurgen, H., Metje, J., Cheema, A., Shafique, N.A., Lawrence, R.D., and Khan, M.A., 1995, Stratigraphic evidence for an early collision between northwest India and Asia: Nature, v. 373, p. 55–58, https://doi.org/10.1038/373055a0.

Besse J., and Courtillot. V., 2002, Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr: Journal of Geophysical Research: Solid Earth, v. 107, 2300, https://doi.org/10.1029/2000JB000050.

Besse, J., Courtillot, V., Pozzi, J.P., Westphal, M., and Zhou, Y.X., 1984, Palaeomagnetic estimates of crustal shortening in the Himalayan thrusts and Zangbo suture: Nature, v. 311, p. 621–626, https://doi.org/10.1038/311621a0.

Bian, W.W., and Yang, T.S., 2021, Reply to Comment by Zhao et al. on "Paleomagnetism of the Late Cretaceous Red Beds from the Far Western Lhasa Terrane: Inclination Discrepancy and Tectonic Implications": Tectonics, v. 40, https://doi.org/10.1029/2020TC006690.

Bian, W.W., Yang, T.S., Ma, Y.M., Jin, J.J., Gao, F., Zhang, S.H., Wu, H.C., and Li, H.Y., 2017, New Early Cretaceous paleomagnetic and geochronological results from the far western Lhasa terrane: Contributions to the Lhasa-Qiangtang collision: Scientific Reports, v. 7, p. 16216, https://doi.org/10.1038/s41598-017-16482-3.

Bian, W.W., Yang, T.S., Jiang, Z.L., Jin, J.J., Gao, F., Wang, S., Peng, W.X., Zhang, S.H., Wu, H.C., Li, H.Y., Cao, L.W., and Deng, C.L., 2020, Paleomagnetism of the Late Cretaceous red beds from the far western Lhasa terrane: Inclination discrepancy and tectonic implications: Tectonics, v. 39, https://doi.org/10.1029/2020TC006280.

Bian, W.W., Yang, T.S., Peng, W.X., Wang, S., Gao, F., Zhang, S.H., Wu, H.C., Li, H.Y., Cao, L.W., Jiang, T., and Wang, H.P., 2021, Paleomagnetic constraints on the India-Asia collision and the size of greater India: Journal of Geophysical Research–Solid Earth, v. 126, https://doi.org/10.1029/2021JB021965.

Cao, Y., Sun, Z.M., Li, H.B., Pei, J.L., Wang, J., Xu, W., Zhao, L.S., Wang, L.Z., Li, C.L., Ye, X.Z., and Zhang, L., 2017, New Late Cretaceous paleomagnetic data from volcanic rocks and red beds from the Lhasa terrane and its implications for the paleolatitude of the southern margin of Asia prior to the collision with India: Gondwana Research, v. 41, p. 337–351, https://doi.org/10.1016/j.gr.2015.11.006.

Chen, W.W., Yang, T.S., Zhang, S.H., Yang, Z.Y., Li, H.Y., Wu, H.C., Zhang, J.H., Ma, Y.M., and Cai, F.L., 2012, Paleomagnetic results from the Early Cretaceous Zenong Group volcanic rocks, Cuoqin, Tibet, and their paleogeographic implications: Gondwana Research, v. 22, p. 461–469, https://doi.org/10.1016/j.gr.2011.07.019.

Ding, L., Maksatbek, S., Cai, F.L., Wang, H.Q., Song, P.P., Ji, W.Q., Xu, Q., Zhang, L.Y., Muhammad, Q., and Upendra, B., 2017, Processes of initial collision and suturing between India and Asia: Science China–Earth Sciences, v. 60, no. 4, p. 635–651, https://doi.org/10.1007/s11430-016-5244-x.

Fisher, R.A., 1953, Dispersion on a sphere: Proceedings of the Royal Society of London, ser. A, v. 217, p. 295–305, https://doi.org/10.1098/rspa.1953.0064.

Hu, X.M., Garzanti, E., Moore, T., and Raffi, I., 2015, Direct stratigraphic dating of India-Asia collision onset at the Selandian (middle Paleocene, 59 ± 1 Ma):




Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

Geology, v. 43, no. 10, p. 859–862, https://doi.org/10.1130/G36872.1.

Hu, X.M., Wang, J.G., BouDagher-Fadel, M., Garzanti, E., and An, W., 2016, New insights into the timing of the India-Asia collision from the Paleogene Quxia and Jialazi formations of the Xigaze forearc basin, south Tibet: Gondwana Research, v. 32, p. 76–92, https://doi.org/10.1016/j.gr.2015.02.007.

Jadoon, U.F., Huang, B.C., Shah, S.A., Rahim, Y., Khan, A.A., and Bibi, A., 2022, Multi-stage India-Asia collision: Paleomagnetic constraints from Hazara-Kashmir syntaxis in the western Himalaya: Geological Society of America Bulletin, v. 134, no. 5–6, p. 1109–1128, https://doi.org/10.1130/B36116.1.

Kapp, P., DeCelles, P.G., Leier, A.L., Fabijanic, J.M., He, S., Pullen, A., Gehrels, G.E., and Ding, L., 2007, The Gangdese retroarc thrust belt revealed: GSA Today, v. 17, no. 7, p. 4–9, https://doi.org/10.1130/GSAT01707A.1.

Kirschvink, J.L., 1980, The least-squares line and plane and the analysis of paleomagnetic data: Geophysical Journal of the Royal Astronomical Society, v. 62, p. 699–718, https://doi.org/10.1111/j.1365-246X.1980.tb02601.x.

Klootwijk, C.T., Conaghan, P.J., Nazirullah, R., and de Jong, K.A., 1994, Further palaeomagnetic data from Chitral (eastern Hindukush): Evidence for an early India-Asia contact: Tectonophysics, v. 237, p. 1–25, https://doi.org/10.1016/0040-1951(94)90156-2.

Kruiver, P.P., Dekkers, M.J., and Heslop, D., 2001, Quantification of magnetic coercivity components by the analysis of acquisition curves of isothermal remanent magnetization: Earth and Planetary Science Letters, v. 189, p. 269–276, https://doi.org/10.1016/S0012-821X(01)00367-3.

Leier, A.L., DeCelles, P.G., Kapp, P., and Ding, L., 2007, The Takena formation of the Lhasa terrane, southern Tibet: The record of a Late Cretaceous retroarc foreland basin: Geological Society of America Bulletin, v. 119, no. 1, p. 31–48, https://doi.org/10.1130/B25974.1.

Li, Z.Y., Ding, L., Song, P.P., Fu, J.J., and Yue, Y.H., 2017, Paleomagnetic constraints on the paleolatitude of the Lhasa terrane during the Early Cretaceous: Implications for the onset of India-Asia collision and latitudinal shortening estimates across Tibet and stable Asia: Gondwana Research, v. 41, p. 352–372, https://doi.org/10.1016/j.gr.2015.05.013.

Liebke, U., Appel, E., Ding, L., Neumann, U., Antolin, B., and Xu, Q., 2010, Position of the Lhasa terrane prior to India-Asia collision derived from palaeomagnetic inclinations of 53 Ma old dykes of the Linzhou Basin: Constraints on the age of collision and post-collisional shortening within the Tibetan Plateau: Geophysical Journal International, v. 182, p. 1199–1215, https://doi.org/10.1111/j.1365-246X.2010.04698.x.

Lin, J.L., and Watts, D.R., 1988, Palaeomagnetic results from the Tibetan Plateau: Philosophical Transactions of the Royal Society of London, ser. A, Mathematical and Physical Sciences, v. 327, p. 239–262, https://doi.org/10.1098/rsta.1988.0128.

Lowrie, W., 1990, Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties: Geophysical Research Letters, v. 17, p. 159–162, https://doi.org/10.1029/GL017i002p00159.

Ludwig, K., 2003, Isoplot/Ex Version 3.00: A Geochronology Toolkit for Microsoft Excel: Berkeley Geochronology Center Special Publication 4, 70 p.

Ma, Y.M., Yang, T.S., Yang, Z.Y., Zhang, S.H., Wu, H.C., Li, H.Y., Li, H.K., Chen, W.W., Zhang, J.H., and Ding, J.K., 2014, Paleomagnetism and U-Pb zircon geochronology of Lower Cretaceous lava flows from the western Lhasa terrane: New constraints on the India-Asia collision process and intracontinental deformation within Asia: Journal of Geophysical Research–Solid Earth, v. 119, p. 7404–7424, https://doi.org/10.1002/2014JB011362.

Ma, Y.M., Yang, T.S., Bian, W.W., Jin, J.J., Zhang, S.H., Wu, H.C., and Li, H.Y., 2016, Early Cretaceous paleomagnetic and geochronologic results from the Tethyan Himalaya: Insights into the Neotethyan paleogeography and the India-Asia collision: Scientific Reports, v. 6, https://doi.org/10.1038/srep21605.

Ma, Y.M., Yang, T.S., Bian, W.W., Jin, J.J., Wang, Q., Zhang, S.H., Wu, H.C., Li, H.Y., Cao, L.W., Yuan, H.F., and Ding, J.K., 2017, Paleomagnetic and geochronologic results of latest Cretaceous lava flows from the Lhasa terrane and their tectonic implications: Journal of Geophysical Research–Solid Earth, v. 122, p. 8786–8809, https://doi.org/10.1002/2017JB014743.

Ma, Y.M., Yang, T.S., Bian, W.W., Jin, J.J., Wang, Q., Zhang, S.H., Wu, H.C., Li, H.L., and Cao, L.W., 2018, A stable southern margin of Asia during the Cretaceous: Paleomagnetic constraints on the Lhasa-Qiangtang collision and maximum width of the Neo-Tethys: Tectonics, v. 37, no. 10, p. 3853–3876, https://doi.org/10.1029/2018TC005143.

Mardia, K.V., and Gadsden, R.J., 1977, A small circle of best fit for spherical data and areas of volcanism: Applied Statistics, v. 26, p. 238–245, https://doi.org/10.2307/2346963.

McFadden, P.L., 1990, A new fold test for paleomagnetic studies: Geophysical Journal International, v. 103, p. 163–169, https://doi.org/10.1111/j.1365-246X.1990.tb01761.x.

McFadden, P.L., and Lowes, F.J., 1981, The discrimination of mean directions drawn from Fisher distributions: Geophysical Journal International, v. 67, no. 1, p. 19–33, https://doi.org/10.1111/j.1365-246X.1981.tb02729.x.

McFadden, P.L., and McElhinny, M.W., 1990, Classification of the reversal test in palaeomagnetism: Geophysical Journal International, v. 103, no. 3, p. 725–729, https://doi.org/10.1111/j.1365-246X.1990.tb05683.x.

Meert, J.G., Pivarunas, A.F., Evans, D.A.D., Pisarevsky, S.A., Pesonen, L.J., Li, Z.X., Elming, S.A., Miller, S.R., Zhang, S.H., and Salminen, J.M., 2020, The magnificent seven: A proposal for modest revision of the Van der Voo (1990) quality index: Tectonophysics, v. 790, https://doi.org/10.1016/j.tecto.2020.228549.

Mo, X., Hou, Z., Niu, Y., Dong, G., Qu, X., Zhao, Z., and Yang, Z., 2007, Mantle contributions to crustal thickening in south Tibet in response to the India-Asia collision: Lithos, v. 96, p. 225–242, https://doi.org/10.1016/j.lithos.2006.10.005.

Parsons, A.J., Hosseini, K., Palin, R.M., and Sigloch, L., 2020, Geological, geophysical and plate kinematic constraints for models of the India-Asia collision and the post-Triassic central Tethys oceans: Earth-Science Reviews, v. 208, https://doi.org/10.1016/j.earscirev.2020.103084.

Pusok, A.E., and Stegman, D.R., 2020, The convergence history of India-Eurasia records multiple subduction dynamic processes: Science Advances, v. 6, https://doi.org/10.1126/sciadv.aaz8681

Rowley, D.B., 1996, Age of initiation of collision between India and Asia: A review of stratigraphic data: Earth and Planetary Science Letters, v. 145, p. 1–13, https://doi.org/10.1016/S0012-821X(96)00201-4.

Sun, Z.M., Pei, J.L., Li, H.B., Xu, W., Jiang, W., Zhu, Z.M., Wang, X.S., and Yang, Z.Y., 2012, Paleomagnetism of Late Cretaceous sediments from southern Tibet: Evidence for the consistent paleolatitudes of the southern margin of Eurasia prior to the collision with India: Gondwana Research, v. 21, p. 53–63, https://doi.org/10.1016/j.gr.2011.08.003.

Tan, X.D., Gilder, S., Kodama, K.P., Jiang, W., Han, Y.L., Zhang, H., Xu, H.H., and Zhou, D., 2010, New paleomagnetic results from the Lhasa terrane: Revised estimation of latitudinal shortening across Tibet and implications for dating the India-Asia collision: Earth and Planetary Science Letters, v. 293, p. 396–404, https://doi.org/10.1016/j.epsl.2010.03.013.

Tang, X.D., Huang, B.C., Yang, L.K., Yi, Z.Y., Qiao, Q.Q., and Chen, L.W., 2013, Paleomagnetism and Ar-Ar geochronology of Cretaceous volcanic rocks in the middle Lhasa terrane, China, and tectonic implications: Chinese Journal of Geophysics, v. 56, no. 1, p. 136–149 [in Chinese with English abstract], https://doi.org/10.6038/cjg20130114.

Tapponnier, P., Mercier, J.L., Proust, F., Andrieux, J., Armijo, R., Bassoullet, J.P., Brunel, M., Burg, J.P., Colchen, M., Dupré, B., Girardeau, J., Marcoux, J., Mascle, G., Matte, P., Nicolas, A., Li, T.D., Xiao, X.C., Chang, C.F., Lin, G.C., Wang, N.W., Chen, G.M., Han, T.L., Wang, X.B., Den, W.M., Zhen, H.X., Sheng, H.B., Cao, Y.G., Zhou, J., and Qiu, Y.H.R., 1981, The Tibetan side of the India-Eurasia collision: Nature, v. 294, p. 405–410, https://doi.org/10.1038/294405a0.

Tapponnier, P., Peltzer, G., Le Dain, A.Y., Armijo, R., and Cobbold, P., 1982, Propagating extrusion tectonics in Asia: New insights from simple experiments with plasticine: Geology, v. 10, p. 611–616, https://doi.org/10.1130/0091-7613(1982)10<611:PETIAN>2.0.CO;2.

Thébault, E., et al., 2015, International Geomagnetic Reference Field: The 12th generation: Earth, Planets, and Space, v. 67, p. 79, https://doi.org/10.1186/s40623-015-0228-9.

Tong, Y.B., Yang, Z.Y., Pei, J.L., Wang, H., Xu, Y.C., and Pu, Z.W., 2017, Paleomagnetism of the Upper Cretaceous red-beds from the eastern edge of the Lhasa terrane: New constraints on the onset of the India-Eurasia collision and latitudinal crustal shortening in southern Eurasia: Gondwana Research, v. 48, p. 86–100, https://doi.org/10.1016/j.gr.2017.04.018.

Tong, Y.B., Yang, Z.Y., Li, J.F., Pei, J.L., and Li, J.F., 2019, New insights into the collision process of India and Eurasia: Evidence from the syntectonic-sedimentation-induced inclinational divergence of Cretaceous paleomagnetic data of the Lhasa terrane: Earth-Science Reviews, v. 190, p. 570–588, https://doi.org/10.1016/j.earscirev.2019.02.009.

Torsvik, T.H., Van der Voo, R., Preeden, U., Niocaill, C.M., Steinberger, B., Doubrovine, P.V., van Hinsbergen, D.J.J., Domeier, M., Gaina, C., Tohver, E., Meert, G.G., McCausland, P.J.A., and Cocks, L.R., 2012, Phanerozoic polar wander, palaeogeography and dynamics: Earth-Science Reviews, v. 114, p. 325–368, https://doi.org/10.1016/j.earscirev.2012.06.007.

van Hinsbergen, D.J.J., Kapp, P., Dupont-Nivet, G., Lippert, P.C., DeCelles, P.G., and Torsvik, T.H., 2011, Restoration of Cenozoic deformation in Asia and the size of Greater India: Tectonics, v. 30, TC5003, https://doi.org/10.1029/2011TC002908.

van Hinsbergen, D.J.J., Lippert, P.C., Dupont-Nivet, G., McQuarrie, N., Doubrovine, P.V., Spakman, W., and Torsvik, T.H., 2012, Greater India Basin hypothesis and a two-stage Cenozoic collision between India and Asia: Proceedings of the National Academy of Sciences of the United States of America, v. 109, p. 7659–7664, https://doi.org/10.1073/pnas.1117262109.

Watson, G.S., and Enkin, R.J., 1993, The fold test in paleomagnetism as a parameter estimation problem: Geophysical Research Letters, v. 20, no. 19, p. 2135–2137 https://doi.org/10.1029/93GL01901.

Yang, T.S., Ma, Y.M., Bian, W.W., Jin, J.J., Zhang, S.H., Wu, H.C., Li, H.Y., Yang, Z.Y., and Ding, J.K., 2015a, Paleomagnetic results from the Early Cretaceous Lakang Formation lavas: Constraints on the paleolatitude of the Tethyan Himalaya and the India-Asia collision: Earth and Planetary Science Letters, v. 428, p. 120–133, https://doi.org/10.1016/j.epsl.2015.07.040.

Yang, T.S., Ma, Y.M., Zhang, S.H., Bian, W.W., Yang, Z.Y., Wu, H.C., Li, H.Y., Chen, W.W., and Ding, J.K., 2015b, New insights into the India-Asia collision process from Cretaceous paleomagnetic and geochronologic results in the Lhasa terrane: Gondwana Research, v. 28, no. 2, p. 625–641, https://doi.org/10.1016/j.gr.2014.06.010.

Yang, T.S., Jin, J.J., Bian, W.W., Ma, Y.M., Gao, F., Peng, W.X., Ding, J.K., Wang, S., Zhang, S.H., Wu, H.C., Li, H.Y., and Yang, Z.Y., 2019, Precollisional latitude of the northern Tethyan Himalaya from the Paleocene redbeds and its implication for Greater India and the India-Asia collision: Journal of Geophysical Research–Solid Earth, v. 124, p. 10,777–10,798, https://doi.org/10.1029/2019JB017927.

Yi, Z., Huang, B., Chen, J., Chen, L., and Wang, H., 2011, Paleomagnetism of early Paleogene marine sediments in southern Tibet, China: Implications to onset of the India-Asia collision and size of Greater India: Earth and Planetary Science Letters, v. 309, p. 153–165, https://doi.org/10.1016/j.epsl.2011.07.001.

Yi, Z., Huang, B., Yang, L., Tang, X., Yan, Y., Qiao, Q., Zhao, J., and Chen, L., 2015, A quasi-linear structure of the southern margin of Eurasia prior to the India-Asia collision: First paleomagnetic constraints from Upper Cretaceous volcanic rocks near the western syntaxis of Tibet: Tectonics, v. 34, p. 1431–1451, https://doi.org/10.1002/2014TC003571.

Yi, Z., Wang, T., Meert, J.G., Zhao, Q., and Liu, Y., 2021, An initial collision of India and Asia in the equatorial



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user

humid belt: Geophysical Research Letters, v. 48, https://doi.org/10.1029/2021GL093408

Yin, A., and Harrison, T., 2000, Geologic evolution of the Himalayan-Tibetan orogeny: Annual Review of Earth and Planetary Sciences, v. 28, p. 211–280, https://doi.org/10.1146/annurev.earth.28.1.211.

Yuan, J., Yang, Z.Y., Deng, C.L., Krijgsman, W., Hu, X.M., Li, S.H., Shen, Z.S., Qin, H.F., An, W., He, H.Y., Ding, L., Gao, Z.T., and Zhu, R.X., 2021, Rapid drift of the Tethyan Himalaya terrane before two-stage India-Asia collision: National Science Review, v. 8, no. 7, p. 1–13 https://doi.org/10.1093/nsr/nwaa173.

Zhao, Q., Huang, B.C., and Tao, W., 2021, Comment on "Paleomagnetism of the Late Cretaceous Red Beds from the Far Western Lhasa Terrane: Inclination Discrepancy and Tectonic Implications" by Bian et al.: Tectonics, v. 40, https://doi.org/10.1029/2020TC006520.

Zhu, D.C., Wang, Q., Chung, S.L., Cawood, P.A., and Zhao, Z.D., 2018, Gangdese magmatism in southern Tibet and India-Asia convergence since 120 Ma, ***in*** Treloar, P.J., and Searle, M.P., eds., Himalayan Tectonics: A Modern Synthesis: Geological Society [London] Special Publication 483, p. 583–604, https://doi.org/10.1144/SP483.14.

Zijderveld, J.D.A., 1967, A.C. demagnetization of rocks: Analysis of results, *in* Collision, D.W., Creer, K.M., and Runcorn, S.K., eds., Methods in Paleomagnetism: Amsterdam, Netherlands, Elsevier, p. 254–286.

Science Editor: Wenjiao Xiao
Associate Editor: Shan Li

Manuscript Received 27 August 2021
Revised Manuscript Received 12 December 2021
Manuscript Accepted 8 February 2022

Printed in the USA



Downloaded from http://pubs.geoscienceworld.org/gsa/gsabulletin/article-pdf/134/11-12/3242/5728094/b36310.1.pdf
by Curtin University Library user